

Existing Credit, New Credit, and Monetary Transmission

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Abstract

Monetary tightening moves the price, the flow, and the stock of revolving credit differently. It reprices credit-card borrowing almost one for one and, under high-frequency identification, weakens the initial limits offered on newly opened accounts. It does not withdraw credit already authorized on open accounts. Across four margins—the limit distribution, line cuts, account exit, and line increases—we find no contraction, and under realized policy-rate changes the estimates reject declines as small as one percent at every observed quantile. A bank holding-company panel reaching back across four tightening cycles to 1990 agrees. Households instead draw more heavily on the capacity they hold: utilization, minimum-payment shares, and delinquency all rise. A short-run accounting framework prices the one channel that stays inactive, the withdrawal of existing lines. Under the baseline calibration, combining the one-percent bound with the framework implies that this stock-quantity channel adds at most 9.5 percent to the modeled short-run spending response to a 100 basis-point tightening. Pre-authorized capacity is economically potent as liquidity but, at the margins we observe, inert to ordinary tightening.

JEL: E52, E21, D14, G51 **Keywords:** monetary transmission, revolving credit, credit limits, household liquidity

1 Introduction

Monetary tightening makes credit-card borrowing more expensive and makes new credit harder to obtain. It does not take back the credit households have already been granted. Those two statements describe different objects, and in revolving credit the second object is by far the larger one. In the median quarter of our sample, newly opened accounts are 3.4 percent of all open credit-card accounts and never more than 4.3 percent, so in the median quarter 96.6 percent of the stock consists of lines authorized in an earlier quarter. Most of that stock sits undrawn: the sample median ratio of unused capacity to the authorized line is 0.950 among cardholders in the household file we build in Section 5, since most cardholders do not revolve. A tightening that operates entirely on new originations leaves that stock untouched, and a line granted before a shock can be drawn precisely when new credit becomes harder or costlier to obtain.

Whether tightening in fact leaves the stock untouched is not obvious, and the presumption runs the other way. Card lines are contractually revocable and are actively managed: issuers raise and cut limits on open accounts using proprietary models, and Bord et al. (2026) document that more than one account in eight receives a limit increase each year. Tightening raises issuers' funding costs and arrives when household default risk is rising, so the incentive to withdraw unused capacity is present. Chava et al. (2023) show that issuers do exactly this: after an adverse wholesale funding shock, banks cut credit limits on existing accounts, and constrained consumers reduce borrowing as a result. That is the same margin, on the same instrument, responding to a different shock. If ordinary monetary tightening operated the same way, the stock and the flow would move together and the distinction above would carry no economic content.

The question is therefore which margin tightening works through. It could restrict revolving credit by repricing borrowing, by tightening the terms on new credit, by withdrawing credit already granted, or by some combination of the three. This paper shows that the stock and the flow do not move together, and bounds what the stable stock could have contributed to short-run spending. We use a quarterly panel of the U.S. large-bank credit-card market that reports, separately and in the same market, the distribution of authorized limits on accounts currently open and the distribution of initial limits on accounts opened in the quarter. That separation is what makes the question answerable: within one market and one specification, the stock and the flow margins of credit supply can be estimated against the same policy shock. We measure policy two ways—realized federal-funds changes, and four high-frequency surprise series—and the two disagree in an informative place.

The price margin. Card borrowing reprices almost immediately. A 100 basis-point increase in the federal funds rate raises the assessed-interest APR by 1.11 percentage points. Most card contracts are variable-rate, so this happens without the issuer revisiting any account's authorized

limit.

The stock margin. Authorized capacity on open accounts does not contract. Capacity already granted can be removed in four distinct ways, and a null on one says nothing about the others, so we test all four. The limit distribution does not shift down: responses are 0.62 percent at the median, -0.15 percent at the seventy-fifth percentile and -0.27 percent at the ninetieth, precise enough to reject contractions as small as one percent at every observed quantile. The share of accounts receiving a line cut does not rise. The accounting-implied rate at which accounts leave the stock does not rise. And the incidence of line increases—the margin through which capacity could stop growing without any limit ever being reduced—does not fall, with declines of 8.2 percent of its sample mean or more rejected. A separate panel of bank holding-company filings covering four tightening cycles, more than twice as long as the card-market panel, gives the same answer for aggregate committed card capacity; it ends in 2018 and so speaks to the cycles through the 2015–2018 normalization rather than to the 2022–2023 tightening.

The flow margin. New-account terms weaken, and here identification decides the answer. Under high-frequency surprises the median initial limit on newly opened accounts falls relative to open-account limits on impact: the difference is negative under all 4 surprise series and under every control set we consider, and is distinguishable from zero for 2 of them. The decline is not a shift in who gets approved, at least as credit scores measure it: the median score of new borrowers does not move, and initial limits fall within score bands. Under realized policy-rate changes the same difference is not identified—it changes sign across control sets and is robustly positive before the pandemic. The contrast is itself informative. In the same data, the count of newly opened accounts rises sharply with realized rate changes and shows no response to any surprise series: realized changes mix the policy impulse with the credit cycle, and the surprises do not. We therefore treat the surprises as the identifying variation for the flow margin and realized changes as the source of economically interpretable magnitudes for the stock.

The flow response rules out the reading that monetary policy simply fails to reach this market: within the same market, quarter, and specification, tightening binds on the terms offered to new borrowers. Its quantity effect appears at origination rather than through the withdrawal of commitments already made, so the finding is the asymmetry rather than either half of it. The asymmetry is across borrowers, not across the direction of policy: estimated separately, a rate cut expands authorized limits no more than a hike contracts them, while the APR passes through in both directions (Section 3.7).

Household use. The capacity does not sit idle. Following tightening, utilization rises by 1.07 percentage points, the share of accounts making only the minimum payment by 0.59 points, and thirty-day delinquency by 0.13 points. With limits broadly stable, higher utilization requires larger balances relative to authorized lines.

Taken together these four margins describe a sequence, and it is not the one a lending-channel account would predict. Monetary policy does not operate on household revolving credit by contemporaneously withdrawing capacity. Borrowing becomes more expensive, the terms on new credit deteriorate, and households with lines authorized before the shock draw on them as cash flow weakens. We call the resulting object *pre-authorized capacity*: credit a household can use without applying for a new account or waiting for a new underwriting decision. What our evidence establishes about it is not that it is a separate argument of household behavior—in the financing identity below it substitutes one for one with liquid wealth—but that it has a different *revocability* profile under tightening: it survives where new credit does not, and it is available at exactly the moment other margins of household adjustment are most constrained.

To quantify what that state implies for short-run spending we build a household accounting framework disciplined by these responses. It isolates the stock-quantity channel: holding the observed price and origination responses common across scenarios, how much larger would the short-run spending response be if tightening also withdrew capacity already authorized? A household enters the quarter with income, liquid assets, an outstanding balance and unused capacity. Tightening lowers cash flow; the household’s expenditure adjusts through an externally calibrated ordinary income response, and realized spending falls further when available financing is insufficient. Pre-authorized capacity relaxes this second, *financing-constraint component* of transmission but not the ordinary income channel. The exercise therefore asks a narrow question: how many households could not finance their observed quarterly expenditure level from current income, liquid assets and unused capacity, and how does that change when capacity is reduced? The framework is a financing identity rather than a structural consumption model; Section 4 states exactly which term of the card channel it maps, and Section 6.5 asks what changes when the higher APR also discourages drawing.

The central quantitative result is a bound on the stock-quantity channel, and its two factors rest on different evidence. The framework supplies the first: holding households’ willingness to draw on an existing line fixed, the modeled short-run spending response rises nearly linearly in the size of a synchronized contraction in authorized lines. The stock-margin estimates supply the second: they bound the contraction that actually occurred at about one percent. Under the baseline calibration their product—the most the stock-quantity channel could have added to the modeled response—is 9.5 percent, a bound rather than an estimate, which a sharper panel would lower without changing the slope. Small cuts bind because unused capacity is thin exactly where it is needed: a given cut in the line is a far larger proportional cut in usable capacity for high-utilization cardholders. The same one-percent cut lowers spending for 5.7 percent of households but for 18.6 percent of cardholders at or above 0.80 utilization, and among affected households the mean loss is \$67 per quarter. Because the cut leaves spending unchanged for approximately nineteen households in twenty, the population-average effect is small for an arithmetic reason rather than an economic one. Section 6 sets out the full ladder and the assumptions that move

it.

What the evidence does not establish. Three limits belong here rather than scattered through the paper. First, the card-market data carry no account identifier and do not report the lower quantiles of the limit distribution, so the stock evidence is distributional: we cannot follow a fixed cohort of accounts through a shock and do not claim that no individual borrower loses capacity, a gap that matters because household-level limit variability is large and consequential even when aggregates are placid (Fulford, 2015). Second, no dataset observes a household’s financing requirement together with its authorized capacity. The framework’s comparison with an economy that has no card capacity is therefore not point-identified, and we report it as a range; the stock-channel bound is much less sensitive to that dependence than to the definition of the financing requirement and to the ordinary marginal propensity to consume. Third, and most consequentially, the exercise holds fixed households’ willingness to draw on an existing line. If a higher APR discourages drawing at the elasticities the literature estimates for revolving balances, the offsetting channel is large enough to reverse the sign of the no-capacity comparison, though not of the stock-channel bound. We therefore do not claim that card capacity attenuates monetary transmission in net terms. Section 6.5 works this out, and Section 7 states what would identify it.

The paper makes three contributions, one for each of its parts.

Our empirical contribution is to show that monetary tightening moves the price, flow and stock margins of revolving consumer credit differently: it reprices existing borrowing, weakens the terms offered at origination, and leaves pre-authorized capacity broadly intact while households draw on it more heavily. Recent work uses account-level and transaction data to measure how policy moves card borrowing costs and spending (Bräuning and Stavins, 2025; Grigoli and Sandri, 2026), studies aggregate unsecured lending in a quantitative monetary model (Littlejohn, 2025), and documents policy effects on new consumer and mortgage lending (Argudo, 2025; Bosshardt et al., 2024; Cucic and Gorea, 2026). We separate, within the same card market and against the same shock, the flow of newly originated credit from the stock of capacity authorized before it, and test the stock at all four margins through which it could be withdrawn. The distinction between limits set at origination and limits revised afterward is not itself new—Santucci (2015) tracks the two separately across the 2005 and 2011 account vintages, and Bord et al. (2026) study the automated limit-increase margin directly. What is new is drawing that distinction against an identified monetary shock, which turns a description of issuer behavior into a statement about transmission. The deposit-pricing literature (Drechsler et al., 2017) explains why the price margin can move without a contemporaneous underwriting decision on an existing account.

Our measurement contribution is to make pre-authorized capacity an observable household state. Unused credit and liquid savings both reduce the sensitivity of spending to income shocks (Baker, 2018); expansions in credit availability raise borrowing and spending (Aydin, 2022;

Gross et al., 2020), with pass-through concentrated among constrained cardholders on data of the same lineage as ours (Agarwal et al., 2018); unsecured credit access shapes adjustment to unemployment and aggregate fluctuations (Herkenhoff, 2019; Braxton et al., 2024); and lenders cut existing lines after adverse funding shocks (Chava et al., 2023). But no single survey records a household’s short-run financing requirement together with its authorized line. We combine three, carrying whole donor rows so that the dependence each survey does observe is preserved, and we report how far the answer moves under the one dependence the construction has to impose. The point is not that unused credit can smooth consumption, which this literature establishes, but that this stock remains available during tightening even as marginal credit conditions worsen.

Our quantitative contribution is to turn the stock null into a bound on a transmission channel, pricing the stability of lines rather than their existence. Balance-sheet positions, marginal propensities, and household debt shape heterogeneous transmission (Auclert, 2019; Kaplan et al., 2018; Cloyne et al., 2020); we add a state whose *revocability* profile differs from that of either liquid wealth or outstanding debt, and ask what the observed absence of a stock response implies. Withdrawing the stock would materially raise financing shortfalls for households near their limits, so the stock matters as liquidity; ordinary tightening does not move it enough for the stock-quantity channel to be large. That credit granted in advance can shield borrowers from later quantity rationing is an old idea for firms: Sofianos et al. (1990) find that monetary policy affects commitment lending less than non-commitment lending, and Woodford (1996) studies how advance loan commitments make policy asymmetric across borrowers. Card lines differ from corporate commitments in the way that matters here. They are revocable and actively managed, so their survival under tightening is an empirical question rather than a contractual guarantee. We show that an economically analogous stock–flow distinction holds for household revolving credit under monetary tightening, and bound what the stock margin could have contributed to consumption.

Section 2 describes the data. Section 3 establishes the seven facts and the stock–flow asymmetry. Section 4 develops the accounting framework and Section 5 maps the household data into its states. Section 6 bounds the stock-quantity channel: Section 6.1 runs the counterfactual, Section 6.2 interprets its magnitude, Section 6.3 reports what moves it, and Sections 6.4 and 6.5 set it against a no-capacity benchmark and APR-sensitive drawing. Section 7 concludes.

2 Data

Our main source is the Federal Reserve Bank of Philadelphia’s *Large Bank Credit Card and Mortgage Data*, a quarterly panel built from the FR Y-14M collection filed by the largest card-issuing institutions. It runs from 2012 to 2025, 54 quarters, and covers the large-bank segment of the general-purpose card market rather than its universe; we describe our estimates accordingly.

The panel reports APRs, utilization, repayment behavior, delinquency, account counts and originations, and the share of open accounts receiving a line increase or a line decrease.

Two features of this source make the paper’s question answerable, and both are unusual. First, it reports the distribution of authorized limits on accounts currently open *and, separately*, the distribution of *initial* limits on accounts opened in the quarter. Most consumer-credit data, including credit bureau files, report one authorized limit per account and cannot distinguish the terms offered to new borrowers from the terms standing on existing ones. That separation is what lets us estimate the stock and flow margins of credit supply within a single market against a single policy shock. Second, it reports the incidence of issuer line increases and decreases directly, so the active management of open accounts can be measured rather than inferred from the limit distribution alone.

Three limitations follow from the same design. The data are market aggregates and carry no account identifier, so the open-account series is a repeated cross-section of the limit distribution, not a tracked cohort of accounts that existed before a particular shock. We therefore use *open-account* throughout for the measured object and reserve *pre-authorized* for its economic interpretation, which the composition figures above support. The panel does not report the tenth or twenty-fifth percentile of the open-account limit distribution, so our distributional statements cover the median and above. And the line-increase series is a share of accounts, not a dollar amount, so it constrains the incidence of forgone increases and not their size.

A second, much longer panel provides an independent check. We measure unused credit-card commitments at bank holding companies from FR Y-9C filings as committed capacity net of drawn balances, covering 70,523 institution-quarters for 1,911 institutions from 1990 to 2018—four tightening cycles, the last of them the 2015–2018 normalization, which the panel covers to 2018Q2 and therefore does not carry to its final hike. Two caveats attach to the item and we return to them in Section 3: it nets drawn balances out of committed capacity, so it moves with borrowing as well as with underwriting, and its definition changes in 2010 from a measure spanning household and commercial card lines to one restricted to consumer lines. We drop differences that span the break rather than splice the series.

The quantitative exercise needs five household states for the same unit: a short-run financing requirement, current income, liquid assets, the authorized card line, and the outstanding balance. No survey measures all five, so we combine three. The Survey of Consumer Finances main public files report authorized limits, balances, liquid assets, and income for the same household, so the dependence among the balance-sheet states is observed rather than assumed; we use the main files rather than the summary extract because only the former report household credit limits directly. The Consumer Expenditure Survey interview files supply the joint distribution of income and total expenditure, which is what we use to measure the financing requirement. The 2024 Diary of Consumer Payment Choice household file supplies the anchors and weights, together with two states that enter the construction directly: card access, which determines

whether a household holds any capacity at all, and revolving status, which is the conditioning variable for drawing balance-sheet donors. Revolving status is economically meaningful rather than a label, because cards serve simultaneously as payment instruments and as revolving debt and convenience use has to be separated from borrowing (Fulford and Schuh, 2023). The household accounting sample contains 5,583 units. The Diary plays no role in the credit-supply results of Section 3, and we make no use of its transaction-level records. Section 5 describes the matching construction and Table A.1 lists sources and coverage; variable definitions and sample restrictions are in the appendix.

2.1 Stock and flow margins

Throughout, we distinguish two credit-supply margins. The *flow margin* is the set of terms assigned to accounts originated after a policy shock, including initial credit limits on newly opened accounts. The *stock margin* is the distribution of authorized limits on accounts open in the quarter. Whether the two respond similarly to tightening is our central empirical question.

For a household with an open account, the relevant liquidity stock is unused revolving capacity,

$$K_i = h_i \max\{A_i - B_i, 0\}, \quad (1)$$

where A_i is the authorized line, B_i the outstanding balance, and h_i an indicator for card access. This is capacity a household can draw without applying for a new account or waiting for a new underwriting decision. Issuers may revise or withdraw limits, and Chava et al. (2023) show they do so after adverse funding shocks; whether they do so under monetary tightening is the question of Section 3.

3 Does Tightening Withdraw Authorized Credit?

This section establishes the empirical pattern the rest of the paper quantifies. We organize it around seven facts: one on the price margin, four on the stock margin, one on the flow margin, and one on how households respond. Table 1 collects the responses; Section 3.6 explains which policy measure identifies which fact.

All magnitudes below are scaled to a 100 basis-point change in the federal funds rate, so it is worth fixing how large that is here. Our sample contains 26 quarters of tightening and 14 of easing in 43 usable quarters, and the 2022–2025 cycle raised the funds rate by 3.82 percentage points on net. A 100 basis-point move is therefore about a quarter of the net increase over that window, and roughly the size of a single brisk year within it.

All responses are estimated by local projection (Jordà, 2005) in first differences, regressing the change in the outcome from $t - 1$ to $t + h$ on the policy measure dated t together with lagged unemployment, the VIX, delinquency, utilization, and a linear trend. Inference is Newey–West with a lag truncation of $\min(h + 1, 8)$ and Student- t critical values at the regression’s residual

degrees of freedom, which are 35 for the limit regressions. Every fact below rests on the impact response, $h = 0$; the appendix reports later horizons, which are weaker evidence throughout and which we do not use to build dynamic profiles.

Table 1: Stock and flow margins of revolving credit after monetary tightening

	Estimate	95% CI
<i>Repricing</i>		
Assessed-interest APR (pp)	1.11	[0.86, 1.37]
<i>Open-account stock</i>		
Open-account limit, $p50$ (%)	0.62	[−0.50, 1.75]
Open-account limit, $p75$ (%)	−0.15	[−0.86, 0.56]
Open-account limit, $p90$ (%)	−0.27	[−0.93, 0.40]
Accounts receiving a line cut (pp)	−0.004	[−0.098, 0.091]
Accounts receiving a line increase (pp)	0.38	[0.02, 0.73]
<i>New-credit flow</i>		
New-account initial limit, $p50$ (%)	−2.77	[−8.20, 2.98]
<i>Use of existing capacity</i>		
Utilization, $p50$ (pp)	1.07	[0.35, 1.80]
Minimum-payment share (pp)	0.59	[0.34, 0.85]
30-day delinquency (pp)	0.13	[0.05, 0.21]

Notes: Responses to a 100 basis-point increase in the federal funds rate. Local projections in first differences on the realized quarterly change in the federal funds rate, with lagged unemployment, VIX, delinquency, utilization, and a linear trend; Newey–West standard errors with lag truncation $\min(h+1, 8)$, so lag 1 at impact, and Student- t critical values at the residual degrees of freedom ($n = 42$, $df = 35$ for the limit regressions). Dollar limits are CPI-deflated and enter in logs, reported as $100(e^\beta - 1)$. The initial limit on newly opened accounts is reported at the median; the ninetieth percentile is imprecise and appears in the appendix. The two line-action rows are shares of open accounts receiving an issuer-initiated limit decrease or increase in the quarter, not the size of those changes. A null on the increase share bounds the *incidence* of forgone line growth, not its dollar magnitude.

3.1 The price margin

Fact 1. Card borrowing reprices nearly one for one with policy. A 100 basis-point increase in the federal funds rate raises the assessed-interest APR by 1.11 percentage points. Most card contracts are variable-rate, with the APR set as a short benchmark rate plus a spread fixed at origination, so policy can raise the price on an existing account without the issuer revisiting that account’s authorized limit (Drechsler et al., 2025).

For scale, the same specification on the 24 quarters from 2015 to 2021, the window over which the national rate on non-jumbo savings deposits is available, gives 0.61 percentage points for the card APR against 0.0177 for the savings rate, a ratio of about 34.5. The cost of drawing on the credit buffer rises by far more than the return to holding the savings buffer.

3.2 The stock margin

Capacity already authorized on an open account can disappear in four distinct ways, and they are not interchangeable. The distribution of authorized limits can shift down; the incidence of

issuer-initiated line cuts can rise; accounts can leave the stock altogether; or the increases that lines would ordinarily receive can be withheld, so that capacity stops growing without any limit ever being reduced. The first three ask whether tightening *withdraws* capacity already granted. The fourth asks whether tightening slows its subsequent *expansion*, and a null on the first three says nothing about it. We test all four.

Fact 2. The open-account limit distribution does not shift down. Responses are small and show no broad downward movement of the distribution. The benchmark estimates are 0.62 percent at the median, -0.15 percent at the seventy-fifth percentile, and -0.27 percent at the ninetieth. The useful question is not whether these differ from zero but how large a contraction the data rule out, so Table 2 reports one-sided tests of economically meaningful negative responses.

Table 2: Economic bounds on contractions in open-account authorized capacity

Margin	Estimate (%)	95% CI	p -value, $H_0 : \beta \leq -c$		
			$c = 1\%$	$c = 2.5\%$	$c = 5\%$
Open-account, $p50$	0.62	$[-0.50, 1.75]$	0.003	<0.001	<0.001
Open-account, $p75$	-0.15	$[-0.86, 0.56]$	0.010	<0.001	<0.001
Open-account, $p90$	-0.27	$[-0.93, 0.40]$	0.015	<0.001	<0.001
New-account, $p50$	-2.77	$[-8.20, 2.98]$	0.736	0.539	0.209

Notes: One-sided tests of the null that the true response is at least as negative as $-c$ percent per 100 basis points. Small p -values reject a contraction of that size. Intervals and p -values use Student- t critical values with the regression’s residual degrees of freedom, the same convention as Table 1 and Table A.2. The card-market data do not report the $p10$ or $p25$ of open-account limits, and carry no account identifier.

At each observed quantile the benchmark estimates reject a contraction as large as one percent, with p -values of 0.0028, 0.0099, and 0.0152. The small stock response is therefore not a consequence of weak statistical power.

That bound depends on the policy measure, and the reason is mechanical rather than substantive. Table A.2 repeats it under each high-frequency surprise, rescaled to 100 basis points. The surprises have standard deviations of 5.52 to 8.50 basis points, so a statement about a 100 basis-point change requires a twelve- to eighteen-fold extrapolation and the intervals widen by the same factor against a fixed one-percent threshold. Of the 12 high-frequency open-account cells, 1 rejects a one-percent contraction; 3 of 3 do so under realized changes. We therefore rest the economic bound on realized rate changes and read it as a statement about power rather than about causal identification.

Three of those cells are significantly negative on impact, and they are not scattered. They are the Jarociński and Karadi (2020) pure-policy response at the seventy-fifth percentile, -0.8 percent per standard deviation, and the Bauer and Swanson (2023) and statement-window responses at the ninetieth. The sign pattern in Table A.2 is monotone in the quantile: no open-account cell is negative at the median under any surprise series, three of the four series

are negative at the seventy-fifth percentile, and all four are at the ninetieth. Read literally, that is a rotation of the open-account limit distribution rather than a downward shift of it—the median authorized line does not fall, the top of the distribution does—and it deserves to be taken seriously rather than dismissed, because unused capacity is concentrated precisely in the upper tail and the mechanism of Section 6 runs through households whose capacity is thin.

Two considerations bound how much weight the pattern can carry. First, these are per-standard-deviation responses to surprises of 5.52 to 8.50 basis points, so the rescaled magnitudes are twelve- to eighteen-fold extrapolations rather than observed movements. Under realized changes—the only measure with variation of the relevant size—the point estimates decline across quantiles in the same direction, but no quantile estimate is distinguishable from zero, and every one of them rejects a one-percent contraction. Second, the largest of the three cells, the Jarociński–Karadi response at $p75$, survives every control set but is indistinguishable from zero at every later horizon ($p = 0.0059$ on impact against a smallest later-horizon p of 0.1394; Table A.11 gives the path).

We therefore state Fact 2 most firmly for the observed median of the open-account limit distribution, and we record the upper tail as unresolved: under high-frequency identification we cannot rule out that issuers trim the largest undrawn lines first. Section 6.3 returns to this, because a rotation that removes capacity from the top of the distribution and a shift that removes it uniformly have different implications for the households the quantitative exercise identifies as constrained.

The 2022–2025 window. One split of the sample matters for interpretation, because it is the only large tightening episode the card panel contains. Estimated on 2022–2025 alone under the full control set, the open-account median response is -3.5 percent per 100 basis points ($p = 0.0114$), against $+4.49$ percent ($p = 0.0464$) in 2012–2019. Read as two independent estimates, the full-sample null of 0.62 percent is a precise average of two opposite regimes, and Table A.3 reports both.

Two checks say that reading does not survive contact with the data. First, each window fits seven parameters on fourteen to nineteen usable observations. Estimating one regression on all 42 quarters instead, with Δi_t interacted with a post-2021 indicator and the control coefficients held common across regimes, gives a pre-2022 response of $+1.48$ percent, an interaction of -1.27 percent, and a 2022–2025 response of $+0.21$ percent. The interaction—which is the direct test of equal policy responses in the two regimes, and the test the subsample comparison cannot provide—has $p = 0.5283$. Second, the 2022–2025 estimate is not robust to the control set: across the six sets of Table A.7 it ranges from -3.55 to $+0.30$ percent, is distinguishable from zero in 4 of 6, and is essentially zero under trend-only and no-controls specifications, while the full-sample estimate is stable and near zero throughout.

We therefore do not read the window as evidence that the stock contracted in 2022–2025. We also do not claim it as support: fourteen observations cannot establish either. What it does

establish is that the sample contains one episode in which the point estimate is negative, and that the paper’s evidence on the stock margin during the largest tightening in its sample is thin.

Fact 3. Line cuts do not become more frequent. The share of open accounts receiving a decrease changes by -0.004 percentage points per 100 basis points, with a confidence interval of $[-0.098, 0.091]$.

Fact 4. Accounts do not leave the stock faster. A stable cross-sectional distribution could in principle coexist with accounts exiting. The data do not report closures, but the accounting identity linking the account stock to new originations implies a net exit measure,

$$\text{exit rate}_t = \frac{\text{new accounts}_t - (\text{accounts}_t - \text{accounts}_{t-1})}{\text{accounts}_{t-1}}, \quad (2)$$

with the beginning-of-quarter account stock in the denominator. We call this the accounting-implied net exit rate and do not read it as literal closures, since it nets entry and exit within the quarter. The series is well behaved, positive in all 53 quarters with a median of 2.94 percent per quarter, and it does not rise after tightening: the realized-rate response is $+0.028$ percentage points (*se* 0.42), and under the four high-frequency surprises every point estimate is negative.

Fact 5. The incidence of line increases does not fall. This is the margin the first three tests miss. Suppose a household enters the quarter with an authorized line of \$10,000 that would have been raised to \$10,500 absent the shock, and under tightening it stays at \$10,000. Nothing has been withdrawn, so Facts 2–4 are silent, and yet capacity is lower than it would otherwise have been. The panel reports the share of open accounts receiving an increase as well as the share receiving a decrease. That share averages 3.04 percent per quarter, roughly twice the line-cut share, so it is the larger of the two active-management margins and the one with more room to move.

Under realized policy-rate changes the response is $+0.38$ percentage points per 100 basis points (*se* 0.17), interval $[0.02, 0.73]$. The point estimate is positive, but we do not read it as evidence that tightening *raises* line increases: dropping calendar 2020, when the line-increase share fell by two thirds as the funds rate was cut to zero, leaves $+0.084$ (*se* 0.11). The reading is the one we give to new-account counts below—realized rate changes mix the policy impulse with the credit cycle. Under all four high-frequency surprises the impact response is economically negligible: 2 of the four point estimates are negative, the number distinguishable from zero is 0, and the largest in absolute value is 0.06 percentage points per standard deviation.

As with the limit distribution, the useful question is how much suppression the data rule out. Table A.5 reports one-sided tests of $H_0 : \beta \leq -c$. A decline of a quarter of a percentage point in the quarterly incidence of line increases—8.2 percent of the sample mean—is rejected at $p = 0.0005$, and larger declines more strongly; the same tests reject at $p = 0.0019$ or below when

calendar 2020 is excluded. Two things bound what that buys. The series is a share of accounts, not an identified household-level hazard, so it constrains the *incidence* of forgone increases and not their size. And the horizon evidence is weak rather than supportive (Table A.6).

The raw series makes the same point without any specification. Between 2022 and 2025, as the federal funds rate rose 3.82 percentage points on net, the median open-account limit rose 13.4 percent in nominal terms and was approximately flat in real terms.

What Facts 2–5 support is a distributional statement, and we hold to it throughout. At the observed margins the open-account limit distribution does not shift downward, the incidence of line cuts and of line increases does not change materially at the tested scale, and accounts do not leave the stock faster. Because the data carry no account identifier, this does not establish that no individual borrower loses capacity, nor that monetary policy leaves incumbent-account management untouched. It establishes that tightening does not broadly withdraw capacity that has already been authorized.

3.3 The flow margin

Fact 6. New-account initial limits fall relative to open-account limits, under surprise identification. The object of interest is the difference between the two responses, not two separately estimated coefficients, so we estimate the benchmark specification directly on $\Delta \log A_{p50}^{new} - \Delta \log A_{p50}^{open}$, which lets inference use the covariance between the outcomes.

Table 3: New-account and open-account credit limits after a monetary-policy surprise

Policy measure	New-account $p50$ (%)	Open-account $p50$ (%)	Difference (%)	95% CI for difference
SF Fed USMPD, market events	−2.34	0.30	−2.64	[−6.01, 0.74]
Bauer–Swanson, orthogonalized	−5.71	0.70	−6.41	[−8.28, −4.54]
Jarociński–Karadi, pure MP	−1.53	0.26	−1.80	[−7.05, 3.46]
SF Fed USMPD, statement	−3.73	0.25	−3.98	[−7.93, −0.03]
<i>Memo</i> : realized Δ fed funds, per 100bp	−2.81	0.62	−3.43	[−8.75, 1.90]

Notes: Rows 1–4 report the response to a one-standard-deviation monetary-policy surprise; surprises are not rescaled to 100 basis points. The difference column is estimated directly on $\Delta \log A^{new} - \Delta \log A^{open}$, so inference uses the covariance between the two outcomes. Limit responses are reported as log points times 100, so the new-minus-open difference remains exactly additive; the single-outcome responses in Table 1 instead use $100(e^\beta - 1)$. Local projections in first differences with lagged unemployment, VIX, delinquency, utilization, and a linear trend; Newey–West standard errors (lag 1 at impact) and Student- t critical values. The memo line uses realized quarterly federal-funds changes and is reported for economic magnitude only; the difference is not identified under that measure, changing sign across control sets (Table A.7).

Table 3 shows that identification decides the answer. Under all 4 surprise series the difference is negative, and for 2 of them the confidence interval excludes zero; the orthogonalized Bauer and Swanson (2023) surprise gives −6.41 percent per standard deviation with an interval of [−8.28, −4.54]. These signs hold under every control set we consider. Under realized federal-

funds changes the same difference is -3.43 percent per 100 basis points with $p = 0.1999$, and it is not identified: dropping the lagged controls flips its sign, and in the 2012–2019 subsample it is robustly positive. Table A.7 reports these permutations.

We read the contrast as informative about what realized rate changes measure. Tightening cycles coincide with credit expansion, and in the same data the count of newly opened accounts rises sharply with realized rate changes while showing no response to any surprise series. Realized changes mix the policy impulse with the credit cycle; the surprises do not.

Two restrictions apply to Fact 6. The first is that the estimates are conditional on origination, so changes in the composition of applicants and approved borrowers may contribute. A fixed difference between new and existing borrowers does not, because the dependent variable is a first difference; the concern is composition that moves with the shock, and it has no presumptive sign. If tightening worsens the pool of applicants, the median initial limit can fall with no change in how issuers set limits for a given borrower. If it instead raises approval standards, the approved pool improves and the median initial limit is pushed up, so the measured decline understates the change in limit setting. Because limits are observed only on approved accounts, what the data record is the net of the two. The account counts above bear on this as well: a recomposition of the approved pool large enough to produce the result would ordinarily also move the number of accounts opened, and under the surprises that number does not respond.

The panel also reports the credit scores of new borrowers, which lets us observe that net directly. The median score at origination—the same quantile of the new-account distribution as the limit in Fact 6—changes by -0.29 points per standard deviation of the Bauer and Swanson (2023) surprise, with an interval of $[-0.82, 0.24]$, and is distinguishable from zero under none of the four series. For scale, at the gradient of initial limits across score bands, composition alone would need a shift of about 4.0 points in the median score to produce the difference in Table 3; the interval excludes a shift of that size even under the steepest gradient we consider. Limits also fall within bands. The median initial limit responds by -4.47 percent for borrowers scoring 660 to 719 and by -3.08 percent for those at 720 or above, both distinguishable from zero, so borrowers in the same score band receive smaller lines. Holding the change in the median score fixed leaves the difference at -5.95 percent, inside the interval of Table 3. Table A.8 reports all four series.

This evidence narrows the composition reading without closing it. The within-band responses are smaller in magnitude than the -5.71 percent response of the pooled median, so recomposition inside a band, which the panel does not resolve, may account for part of the pooled decline. And because the panel records only approved accounts, selection at the application stage is not observed. We therefore describe the result as a deterioration in new-account terms rather than as a structural supply elasticity.

The second restriction is that the dynamics are weaker evidence than the impact estimate. Table A.10 reports the difference at horizons zero through four for each series: it is negative

in 12 of the 12 cells through two quarters and distinguishable from zero on impact for 2 of four, but at longer horizons the point estimates are non-monotone, with precision weakening at intermediate horizons and returning for some series at $h = 4$. Nor do we claim the whole new-account distribution shifts down: the ninetieth-percentile estimates are positive and imprecise under realized changes and negative under the surprises, and our formal contrast is centered on the median. Table A.9 reports both sets.

3.4 Household use

Fact 7. Households use existing capacity more intensively. Tightening is followed by higher utilization, which rises by 1.07 percentage points at the median, a larger minimum-payment share, up by 0.59 points, and higher thirty-day delinquency, up by 0.13 points. With limits broadly stable, higher utilization requires households to carry larger balances relative to their authorized lines, consistent with households drawing more heavily on previously granted capacity as cash flow weakens.

We do not read utilization and delinquency as clean causal measures of credit demand in isolation. Their role is to complete the joint pattern: the stock of open-account capacity remains available, and households make greater use of it.

3.5 An independent panel over four tightening cycles

The card-market panel covers 54 quarters. The FR Y-9C filings let us ask the same question of aggregate committed capacity over four tightening cycles, using up to 67,536 institution-quarters and 110 quarters. The two panels are complements rather than substitutes in coverage: the Y-9C panel reaches back to 1990 but ends in 2018Q2, so it adds the three tightening cycles before the card panel begins and adds nothing to the 2022–2023 episode, where the card panel is the only evidence we have.

Of the 18 estimates in Table 4, 1 is distinguishable from zero at the ten percent level, and it is positive. Applying the same one-sided bound as in Table 2, the realized-rate impact estimate rejects a contraction of one percent in committed capacity ($p = 0.0127$). Two-way clustering absorbs within-quarter cross-sectional correlation but not serial correlation in the time dimension, which matters here because the policy regressor varies only in t and the $h = 2$ and $h = 4$ windows overlap by construction. Recomputing the impact row with Driscoll–Kraay standard errors, which are robust to both, inflates the impact standard error at lag eight by 23.6 percent and leaves the bound intact: the one-sided p -value is 0.0336 at a Bartlett lag of four and 0.0353 at eight. At $h = 4$ it does not survive beyond a lag of two. The dependent variable is log commitments at $t + h$ minus log commitments at $t - 1$, and an institution-quarter enters only when every quarter from $t - 1$ to $t + h$ is present and calendar-adjacent, so the horizon is a fixed calendar distance rather than a count of available filings.

Two features of the reporting item bound how far this reaches. It is not a limit elasticity: the

Table 4: Unused credit-card commitments at bank holding companies, 1990–2018

Policy measure	$h = 0$	$h = 2$	$h = 4$
Realized Δ fed funds (per 100bp)	0.03 (0.46)	0.33 (0.80)	0.76 (1.27)
SF Fed USMPD, market events	0.35 (0.17)	0.13 (0.34)	0.54 (0.40)
Bauer–Swanson, orthogonalized	0.01 (0.22)	−0.32 (0.33)	−0.45 (0.40)
Jarociński–Karadi, pure MP	0.11 (0.19)	−0.08 (0.30)	0.01 (0.36)
Miranda–Agrippino–Ricco	−0.04 (0.19)	0.03 (0.27)	0.13 (0.28)
Wu–Xia	−0.07 (0.23)	0.09 (0.36)	0.25 (0.50)
<i>Realized Δ fed funds, alternative inference</i>			
Driscoll–Kraay, lag 2	0.03 (0.53)	0.33 (0.78)	0.76 (1.35)
Driscoll–Kraay, lag 4	0.03 (0.57)	0.33 (0.84)	0.76 (1.46)
Driscoll–Kraay, lag 8	0.03 (0.57)	0.33 (0.84)	0.76 (1.43)

Notes: Percent response of unused credit-card commitments, measured as committed capacity net of drawn balances. Institution fixed effects; standard errors two-way clustered by institution and quarter, in parentheses. Up to 67,536 institution-quarters, 1,911 institutions, and 110 quarters, covering 1990Q3–2018Q2. The dependent variable is log commitments at $t + h$ less $t - 1$, and an observation is used only when every quarter from $t - 1$ to $t + h$ is present and calendar-adjacent, so the horizon is a fixed calendar distance. The policy regressor varies only in t : two-way clustering absorbs within-quarter cross-sectional correlation but not serial correlation in the time dimension, and the $h = 2$ and $h = 4$ windows overlap by construction, so these standard errors may understate sampling uncertainty. Windows spanning the BHCK3815/BHCKJ455 redefinition are dropped. High-frequency rows are per one standard deviation. Of the 18 estimates, 1 is distinguishable from zero at the ten percent level and it is positive. The Driscoll–Kraay rows recompute the realized-rate row with standard errors that are robust to cross-sectional dependence *and* to serial correlation in the time dimension, with a Bartlett kernel at the stated lag; the point estimates are unchanged by construction. Inflation relative to two-way clustering is 23.6 percent at the impact horizon with lag eight. The one-percent bound survives: the one-sided p -value against $H_0 : \beta \leq -1\%$ at $h = 0$ is 0.0264, 0.0336 and 0.0353 at lags 2, 4 and 8, against 0.0127 under two-way clustering. At $h = 4$ it does not survive beyond lag 2.

item nets drawn balances out of committed capacity, so it moves with borrowing as well as with underwriting. That contamination has a known sign here, and it runs against us rather than for us. Fact 7 finds that households draw more heavily on existing lines when policy tightens, so drawn balances rise; since the item subtracts them, the measured series understates committed capacity by more after a tightening than before. A null on this measure is therefore conservative: the same underwriting behavior would show up as a contraction in the reported item before it would show up as a stable one. And it is not a pure consumer-card measure throughout, since the pre-2010 item covers card lines extended for household *and* for commercial or industrial purposes while only the post-2010 item is restricted to consumer lines. Since 58,554 of the 67,536 institution-quarters come from the earlier item, the long panel measures committed card capacity broadly defined. Table A.12 estimates the two regimes separately; neither distinguishes the response from zero on its own, and the post-2010 consumer-only subsample has 8,982 observations and is too imprecise to sharpen the conclusion. What it delivers is evidence that aggregate committed capacity does not collapse during tightening, on a sample more than twice as long as the card-market panel and with a bank-level panel dimension.

Taken together the price, stock, flow, and use margins imply

$$MP \uparrow \implies APR \uparrow, \quad \text{new-account terms weaken,} \quad A^{\text{open}} \approx \text{stable}, \quad B/A \uparrow.$$

The first two responses are active channels of transmission through revolving credit, a price channel and a flow-quantity channel at origination. The third, the stock-quantity channel, is the one the estimates show is largely inactive, and it is the only one the framework of Section 4 quantifies.

3.6 What identifies what

We use two policy measures for two different purposes and do not treat them as interchangeable. Realized federal-funds changes deliver economically interpretable 100 basis-point magnitudes, and the equivalence bounds of Facts 2 and 5 rest on them. High-frequency surprises (Bauer and Swanson, 2023; Jarociński and Karadi, 2020) deliver a more exogenous policy measure, and Fact 6 rests on them. The division is forced by the data rather than chosen: typical surprises are only several basis points, so scaling them to 100 basis points extrapolates well beyond the support of the series and yields equivalence bounds too wide to be informative at the one-percent scale, while realized changes demonstrably mix the policy impulse with the credit cycle in the flow series. We therefore label realized-rate estimates as pass-through magnitudes rather than as fully exogenous causal effects, and use the high-frequency estimates for sign and robustness rather than for magnitude. Under the alternative high-frequency measures, median new-account initial-limit estimates remain negative with precision varying across series, and open-account limit responses remain small. Table A.9 reports the full set of impact responses and Tables A.10

and A.11 the horizon paths.

3.7 Does easing do the reverse?

A null on authorized capacity under tightening invites one further reading: that limits are simply not managed at the quarterly frequency, in which case the result says little about policy. If issuers do manage the stock, easing should expand it. We therefore split the realized policy change by sign and estimate

$$\Delta y_t = \alpha + \beta_T T_t + \beta_E E_t + \Gamma X_{t-1} + \varepsilon_t, \quad T_t = \max\{\Delta i_t, 0\}, \quad E_t = \max\{-\Delta i_t, 0\}, \quad (3)$$

so that β_T is the response to a 100 basis-point hike and β_E the response to a 100 basis-point cut, and a symmetric linear response implies $\beta_T + \beta_E = 0$. Controls, sample convention and inference follow Table 1, and collapsing T_t and E_t back to Δi_t reproduces it exactly. The sample contains 26 tightening and 14 easing quarters for the repricing regression and 26 and 13 for the limit regressions, which run on a series missing one quarter.

The price margin is symmetric. Excluding calendar 2020, a hike raises the assessed-interest APR by 1.29 percentage points, with a confidence interval of $[0.78, 1.80]$, and the cut coefficient is -1.20 points, $[-2.06, -0.34]$; symmetry cannot be rejected ($p = 0.8705$). Repricing runs in both directions at close to the same rate.

The stock margin moves in neither. At the median, open-account limits respond by 0.62 percent to a hike and -0.61 percent to a cut; across the three quantiles all six coefficients are below one percent in absolute value. None is distinguishable from zero, symmetry is rejected at no quantile, and the joint null $\beta_T = \beta_E = 0$ survives at all three ($p = 0.5485, 0.9133$ and 0.6209). The easing side also supports the equivalence bounds that Fact 2 rests on: an expansion of authorized limits of 2.5 percent or more per 100 basis-point cut is rejected at every quantile ($p = 0.0165, 0.0273$ and 0.0157). Table A.13 collects both panels and the bounds.

The asymmetry this paper documents is therefore between existing and new borrowers, not between the two directions of policy. A line already authorized is not rewritten when the rate moves either way, which is what the stock-flow reading implies and what a story about issuers actively managing capacity against the policy rate does not.

We read the easing coefficients as bounds rather than as point estimates. Of the 14 easing quarters, calendar 2020 alone supplies 41.6 percent of the easing variation, so E_t partly measures the pandemic recession rather than monetary easing. Outside 2020 the median open-account limit carries a *negative* easing coefficient (-2.44 percent, $p = 0.0573$), the wrong sign for a monetary interpretation, exactly as the line-increase share does in Table A.4. What the easing side can support is the bound—no economically large expansion of authorized capacity—rather than the sign of the point estimate.

4 A Short-Run Household Accounting Framework

The empirical results motivate a short-run accounting framework in which pre-authorized capacity is a household state rather than a contemporaneous policy choice. We call it a framework rather than a model deliberately. Equation (5) below is a financing identity: it says what a household can spend given its resources and the cap on its borrowing, and it contains no preference parameter and no optimization that the data could refute. It is not a structural model of consumption, and it is not a model of the credit-card channel of monetary policy as a whole. Its purpose is to carry the observed stock of credit capacity into a statement about short-run spending, not to reproduce the dynamic consumption response to monetary policy and not to identify household preferences. Everything the exercise adds beyond arithmetic is in the calibration of the marginal propensity m and the income path δ_y , both taken from outside this paper, and in the cross-survey construction of the household states, which Section 5 describes and Section 6.3 stress-tests.

The framework is also narrower than a model of the card market. Section 3 identifies three card-market responses: a price response, a flow response at origination, and a near-zero stock response on open accounts. Leaving aside the ordinary income channel, which operates whether or not a household holds a card, the card-market part of the spending response can be written schematically as

$$\frac{dC}{dMP} \approx \underbrace{\frac{\partial C}{\partial r} \frac{dr}{dMP}}_{\text{price}} + \underbrace{\frac{\partial C}{\partial A^{\text{new}}} \frac{dA^{\text{new}}}{dMP}}_{\text{flow}} + \underbrace{\frac{\partial C}{\partial A^{\text{open}}} \frac{dA^{\text{open}}}{dMP}}_{\text{stock}}, \quad (4)$$

where r is the card APR. This is a conceptual decomposition rather than an exact identity, since the channels can interact, and the paper has different information about each term. It estimates dr/dMP (Fact 1) but not how spending responds to the APR. It estimates the response of new-account terms (Fact 6) but not how spending responds to them, and those terms are observed conditional on origination. It tightly bounds dA^{open}/dMP (Facts 2–5). The framework below maps only the stock-quantity margin into household financing capacity, so that its sensitivity to a withdrawal of existing lines can be combined with that bound. The APR and new-account responses are common to the stable-stock and line-cut scenarios, and so are not inputs to the baseline stock comparison; Section 6.5 separately asks how allowing the APR increase to discourage drawing changes the result. We do not add a block for new credit: it would require application and approval probabilities and a spending response to origination terms, none of which our data identify, so Fact 6 remains an empirical result about the flow margin rather than a calibration target.

4.1 Household state and financing rule

A household enters the quarter with current income y_i , beginning-of-period liquid assets X_i , an authorized card limit A_i , an outstanding balance B_i , and card access h_i . Its unused revolving capacity K_i is given by Equation (1). Four resources are kept distinct throughout: current income, the liquid buffer, new card borrowing, and the authorized limit that caps it.

Spending follows a hard-capacity rule: the household meets a short-run financing requirement R_i until financing is exhausted. Baseline spending is then

$$c_{i,0} = \min\{R_i, y_i + \lambda_x X_i + K_i\}. \quad (5)$$

We call R_i a *financing requirement* rather than desired consumption, and the distinction is substantive. We measure R_i with observed CEX expenditure, which is an equilibrium outcome: it already reflects the household’s liquid assets, card access, borrowing constraints, debt service, and past shocks. It is not a latent unconstrained optimum. The exercise therefore asks a narrower and more answerable question—how many households could not finance their observed quarterly expenditure level without unused capacity—rather than how far latent desired consumption is suppressed. The appendix states the reduced-form borrowing-cost specification that Section 6.5 uses to relax the hard cap. No preference parameter is estimated. We set $\lambda_x = 1$ because the asset measure is constructed to capture liquid balances, and report lower values as sensitivity. There is no separate parameter on K_i : one dollar of unused authorized credit is at most one dollar of financing capacity. Any reluctance to borrow below the line is treated in Section 6.5.

4.2 The policy experiment and the decomposition

A monetary tightening lowers household cash flow by $\Delta y_i = \delta_y y_i$ and raises the card APR. Under the hard-capacity rule the higher APR does not change what a household can finance, so it does not appear below; it enters only when Section 6.5 lets it discourage drawing. Post-shock consumption is

$$c_{i,1} = \min\{R_i - m_i \Delta y_i, y_i - \Delta y_i + \lambda_x X_i + K'_i\}. \quad (6)$$

The first term is the financing requirement after the household’s ordinary response to lost income; the second is what it can finance from current income, usable liquid wealth, and open-account capacity. Spending falls for two conceptually distinct reasons. Households reduce their requirement even when financing remains available, and some are forced to cut further because their resources are insufficient. Write $\Delta C_i = c_{i,1} - c_{i,0}$ and $\Delta C_i^{\text{ord}} = -m_i \Delta y_i$. Within a pairing, let ΔC^{total} and ΔC^{ord} denote the Diary-weighted means of these household quantities and let ΔC^{con} be the residual:

$$\Delta C^{\text{total}} = \Delta C^{\text{ord}} + \Delta C^{\text{con}}. \quad (7)$$

This decomposition is central to interpretation. The ordinary component does not involve

K_i and is present whether or not the household holds a card. Pre-authorized capacity affects only the financing-constraint component. The framework therefore does not predict that credit cards weaken the ordinary income channel of monetary policy.

4.3 Calibration of m_i and δ_y

We set the quarterly ordinary MPC to $m = 0.25$ in the baseline. This is an external, literature-disciplined calibration, not a parameter estimated from our data. U.S. quasi-experimental evidence on temporary cash-flow shocks places short-horizon spending responses near this value: Johnson et al. (2006) estimate that households spent roughly 20 to 40 percent of the 2001 rebates on nondurables within the quarter of receipt; Parker et al. (2013) obtain 12 to 30 percent for the 2008 payments; and Baker et al. (2023), using high-frequency transaction data, find a cumulative response near 25 to 30 cents per dollar over three months. We report $m = 0.15$ and $m = 0.40$ as lower and upper cases and a coarse liquid-wealth schedule as further robustness.

Two features of this mapping matter. Most clean U.S. evidence concerns positive shocks, whereas our experiment is an income loss, and survey evidence suggests responses can be larger for losses than for gains (Fuster et al., 2021). Because that evidence is survey-based or non-U.S., we do not raise the baseline; the upper-MPC case spans this uncertainty. Separately, we deliberately use a common m rather than a liquid-wealth-graded schedule in the baseline. Part of the high measured MPC of low-liquidity households reflects binding financing constraints, which our framework generates explicitly through K_i ; assigning that behavior to m_i as well would double-count the same mechanism.

We set $\delta_y = 0.54$ percent per 100 basis points from Furceri et al. (2019), and its horizon matters. They estimate the cumulative response of state-level real personal income to an exogenous 100 basis-point policy-rate increase in a spatial local projection. In their baseline specification that cumulative response is 0.81 percent at the fourth quarter; our δ_y is the mean of the cumulative responses over the first four quarters, which is the relevant object for a single-quarter accounting experiment rather than the terminal value. Two features of the mapping are approximations we do not hide: the outcome is state personal income rather than household disposable income, and the shock is identified on a 1969–2008 sample. δ_y is therefore a scale calibration, not a matched moment; it scales the dollar response in both arms of every comparison below, and Appendix B.1 shows the attenuation ratio is nearly unchanged across a twelvefold range of it.

4.4 The credit response in the counterfactual

The baseline holds open-account authorized lines fixed on impact, $A'_i = A_i$, consistent with Facts 2–5. This does not assume card limits are technologically fixed; it normalizes the impact response of open-account authorized lines to zero, and Section 6.3 passes the estimated point response and its confidence bounds through the same block. The policy-aligned counterfactual,

developed in Section 6, instead applies a proportional contraction to the authorized line,

$$A'_i = (1 - \kappa)A_i, \quad K'_i = \max\{(1 - \kappa)A_i - B_i, 0\}, \quad (8)$$

treating the outstanding balance as predetermined over the impact period. The mapping in Equation (8) matters because the empirical estimates concern authorized limits A_i , not unused capacity K_i . Removing capacity entirely, $K_i = 0$, is retained as an extreme benchmark rather than as the central comparison.

The line-cut scenario is not a prediction that issuers cut lines; Facts 2–5 say they do not. It measures the stock term of Equation (4)—how much spending adjustment a given movement in existing authorized lines would generate—so that this sensitivity can be combined with the empirical bound on how far those lines actually moved.

Table 5 lists every input the framework uses, what disciplines it, and its status. The last column states the direction in which each input moves the headline amplification, so that the reported figure can be read against the assumptions that produce it rather than only against its own standard error.

5 Mapping Households into the Framework

5.1 Household states

The framework needs five household states for the same unit: the financing requirement R_i , current income y_i , liquid assets X_i , the authorized card line A_i , and the outstanding balance B_i . No single survey measures all five. The quantitative sample is anchored on the 5,583 households in the 2024 Diary household file, which supplies the weights, the income group, card access, and revolving status. The CEX supplies (y_i, R_i) and the SCF supplies (X_i, A_i, B_i) , each carried as a whole row from a single donor household so that the dependence a survey does observe is preserved. CEX financing requirements are assigned by weighted donor draw within income cells. The objective is not a synthetic panel with unobserved precision; it is to reproduce the distributions that enter Equations (5) and (6) as closely as the separate surveys allow.

Income and the financing requirement are carried jointly from the same CEX donor, preserving their observed joint distribution rather than assembling it from two marginals. The balance-sheet block follows the same rule: liquid assets, the authorized line, and the balance are carried jointly from a single SCF donor drawn within income-group $\times$ revolver cells, and utilization and unused capacity follow as $u_i = B_i/A_i$ and $K_i = h_i \max\{A_i - B_i, 0\}$ as in Equation (1). The SCF observes liquid assets, the authorized line, and the balance for the same household, so their dependence is preserved rather than assumed.

The two donor blocks are then paired *conditionally randomly* within the common matching cells. That pairing is the only dependence in the construction that is imposed rather than

Table 5: Inputs to the accounting framework

Object	Discipline	Status	Direction of effect on the reported amplification
R_i, y_i : financing requirement and income	CEX donor block	Observed	Baseline uses total expenditure, the top of the 1.3–9.5 percent range; narrower expenditure concepts lower it
X_i, A_i, B_i : liquid assets, authorized line, balance	SCF donor block	Observed jointly	Entered as a joint distribution, not margin by margin
Card access, revolver status	Diary anchor	Observed	Fixes which households have a line at all
Dependence between R_i and the balance-sheet block	Conditional random pairing	Assumed; swept	7.2–9.3 percent across the dependence constructions reported
$\lambda_x = 1$: usable share of liquid assets	Normalization	Assumed	Lowering it raises the amplification, so $\lambda_x = 1$ is the conservative end
$m = 0.25$: ordinary expenditure response	External MPC evidence	Calibrated	14.3 percent at $m = 0.15$, 6.3 percent at $m = 0.40$; higher m lowers it
$\delta_y = 0.54$ percent per 100 bp	External monetary-policy evidence	Calibrated	Scales both arms; the attenuation ratio is nearly invariant over a twelvefold range
κ : authorized-line contraction	Scenario, bounded by Facts 2–5	Counterfactual	Linear in κ at 9.6 percent per percentage point over the relevant range
d : APR-sensitive drawing	External card-elasticity evidence	Not identified here	Falls to 4.6 percent at the long-run anchor

Notes: *Observed* means taken from survey data without a behavioral assumption; *assumed* means imposed by the construction and varied in sensitivity; *calibrated* means set from evidence external to this paper and not matched to any moment of our data; *counterfactual* means the object the policy experiment varies. The headline amplification is 9.5 percent for a one-percentage-point synchronized contraction and measures the stock-quantity channel only; neither the APR response of Fact 1 nor the new-account response of Fact 6 enters it. No entry in this table is estimated from the variation used in Section 3.

observed, and we state it as an assumption: conditional on the cell, no additional rank dependence is placed between the financing requirement and the balance-sheet vector. Section 6.3 and Appendix B.2 vary exactly that dependence and report how far the answer moves.

Because the pairing is random, a single draw is not the object of interest. Fixing the SCF block and redrawing the pairing moves the attenuation of the financing-constraint component with a standard deviation of about three percentage points, so any one seed would put a headline number in the hands of the random number generator. All headline household-accounting aggregates are therefore means over 50 independent cross-survey pairings, with the Monte Carlo standard deviation carried in the output; Table 6 and Figure 1 use one representative pairing, and the companion diagnostic in Table B.4 is averaged over ten simulations.

That pairing error is not the only statistical uncertainty in the block, and it is one of three components we can compute. Table B.11 decomposes it. The second source is the Board’s imputation of the SCF: the donor pool pools all five implicates, so the point estimate already averages over them, but restricting the pool to one implicate at a time moves the amplification with a standard deviation of 0.17 percentage points and the attenuation of the constraint component by 1.00. The third, and the largest for most statistics, is sampling error in the CEX itself, which enters through the donor-selection probabilities; the 44 replicate weights supplied with the Interview survey give a balanced-repeated-replication standard error of 0.38 percentage points on the amplification. Added in quadrature the three give 0.45 percentage points on an amplification of 9.5 percent and 1.22 on an attenuation of 39.2 percent.

The relative magnitudes depend on the statistic. The combined standard error on amplification is 0.45 percentage points, compared with a range of 7.2 to 9.3 percent across the reported dependence constructions and 6.3 to 14.3 percent across MPC calibrations. For constraint attenuation, the combined standard error is 1.22 percentage points, compared with 23.4 to 61.8 percent across the central capacity-dependence sweep. These comparisons motivate organizing Section 6 around the assumptions, without treating every sensitivity range as an order of magnitude larger than statistical uncertainty. We do not apply the Diary or SCF replicate weights and treat the three components as independent, so the totals in Table B.11 do not cover all sampling uncertainty.

Table B.5 shows what the joint-donor rule buys. The natural alternative assigns liquid assets, the line, and utilization from three separate conditional distributions, each ordered within cells by its own economic score. That targets the separate marginal distributions but replaces the dependence the SCF observes: within income groups the average absolute deviation from the survey rank correlation is 0.33, and the positive association between liquid assets and authorized capacity comes out with the right sign in only 33.3 percent of income groups. The joint donor cuts that deviation to 0.03 and reproduces the observed sign in every cell. Table B.6 reports the score-based construction as an alternative assignment, and Table 6 the resulting states for 5,583 synthetic households.

Table 6: Household states by income

Household income	Card access	Unused capacity K ($p50$)	Liquid assets X ($p50$)	Income y ($p50$)	Financing constrained
Under 25,000	47.3%	\$2,679	\$978	\$3,623	47.8%
25,000–49,999	74.2%	\$6,696	\$2,644	\$9,047	21.3%
50,000–74,999	84.8%	\$6,837	\$5,789	\$15,371	9.0%
75,000–99,999	87.1%	\$12,656	\$10,291	\$21,434	2.6%
100,000–149,999	93.8%	\$18,852	\$12,825	\$30,761	2.4%
150,000+	97.7%	\$44,196	\$48,921	\$53,248	0.8%

Notes: 5,583 synthetic households, shown for one representative donor pairing; the aggregate counterfactuals below are means over 50 independent cross-survey pairings. Unused capacity is conditional on card access; income is a quarterly flow; liquid assets are a beginning-of-period stock. Both are weighted medians. A household is financing constrained when its financing requirement exceeds available resources, $R > y + \lambda_x X + K$, with $\lambda_x = 1$.

Two features of Table 6 drive the results below. Card access rises from 47.3 percent of households below \$25,000 to 97.7 percent above \$150,000, and median unused capacity among cardholders rises from \$2,679 to \$44,196. Pre-authorized capacity is therefore concentrated where liquid wealth is already ample, which is what limits how much of aggregate transmission it can absorb.

6 Quantitative Implications of Stock Stability

This section quantifies the stock term of Equation (4), the stock-quantity channel that the estimates of Section 3 show is largely inactive. The price and flow terms remain outside the exercise, for the reasons given in Section 4.

6.1 What if open-account limits had tightened too?

The counterfactual asks how much a withdrawal of authorized lines would have added to the short-run spending response, given that the estimates find none. It applies Equation (8), cutting the authorized line in the post-shock period with the balance predetermined, rather than a proportional cut in K_i itself, because Facts 2–5 concern the distribution of authorized lines. The two are far apart where it counts: the sample median ratio of unused capacity to the authorized line is 0.950 among cardholders but only 0.091 among high-utilization cardholders, so a given cut in A implies a much larger proportional cut in K for the households for whom unused capacity is thinnest. Table 7 reports the result on the same Diary anchors and donor rules as Table 6, as means over 50 cross-survey pairings.

The response is nearly linear in the size of the cut. Fitted through the rungs, the amplification is 9.6 percent per percentage point of synchronized withdrawal, and no rung departs from that line by more than 1.3 percent over cuts between one and five percent. Each percentage point of authorized capacity withdrawn from open accounts alongside a 100 basis-point tightening

Table 7: What if open-account authorized lines had tightened too?

Scenario	ΔA	ΔK		Constraint component	ΔC total	Ampli- fication	Share of ladder
		mean	$u \geq 0.8$				
Stable stock	—	0	0	−4.70	−39.87	—	—
Line −1%	−1.0%	−1.1%	−8.3%	−8.49	−43.66	+9.5%	0.8%
Line −2.5%	−2.5%	−2.8%	−20.4%	−14.23	−49.40	+23.9%	2.0%
Illustrative −2.77%	−2.8%	−3.1%	−22.4%	−15.27	−50.44	+26.5%	2.3%
Line −5%	−5.0%	−5.7%	−38.7%	−23.96	−59.12	+48.3%	4.1%
Full withdrawal, $\kappa = 1$	−100%	−100%	−100%	−475.37	−510.54	+1180.5%	100.0%
<i>Memo: no existing capacity</i>	—	—	—	−7.73	−42.90	+7.6%	—

Notes: Quarterly dollars per household at $m = 0.25$, computed on the same 5,583 Diary anchors and preferred donor rules as Tables B.1 and 6; aggregates are means over 50 independent cross-survey pairings. Each scenario cuts the *authorized line* in the post-shock period, $A'_i = (1 - \kappa)A_i$, with the outstanding balance predetermined, so unused capacity becomes $K'_i = \max\{(1 - \kappa)A_i - B_i, 0\}$. Amplification is $|\Delta C^\kappa|/|\Delta C^{\text{stable}}| - 1$. *Share of ladder* expresses each rung’s increment as a fraction of the increment at the top rung, $\kappa = 1$ (full withdrawal of the authorized line after the shock, with the balance still predetermined). That benchmark shares the pre-shock state $c_{i,0}$ with every other row, so the column is bounded by 100 percent and is invariant to whether responses are expressed in dollars or as shares of each arm’s own baseline spending. The memo row is the *different* comparison used in Section 6: an economy with no card capacity in *both* periods. It is a two-economy comparison, not a policy derivative—its pre-shock spending is \$17,820 against \$18,288 here—which is why it is reported as a memo rather than as the ladder’s denominator. *Aggregation.* The amplification and share columns are means of per-draw ratios, $E[X/Y]$, not ratios of the reported means, $E[X]/E[Y]$. For amplification the two agree to 0.033 percentage points across the partial-cut rows, so recomputation from the ΔC means is approximate. The full-withdrawal amplification is the ratio of the reported dollar means. For the share column against the old no-capacity denominator the two differed materially and grew apart with κ (126.45 versus 124.96 percent at $\kappa = 0.01$); against the $\kappa = 1$ denominator used here they agree to 0.018 percentage points. The −2.77 percent row is calibrated to the magnitude of the median new-account point estimate and is illustrative, not an estimated issuer response for open accounts.

therefore raises the modeled short-run spending response by about 9.6 percent. Linearity is a property of this range and not of the mapping in general—full withdrawal, $\kappa = 1$, raises the response by 1180.5 percent, far above the fitted line—but it covers every cut the paper’s evidence makes relevant.

The slope and the size of the cut are the two factors of the stock term in Equation (4), and they rest on different evidence. The slope is the accounting counterpart of $\partial C / \partial A^{\text{open}}$, a property of the framework and the household states that feed it; Section 6.3 reports what moves it. The size of the cut is dA^{open} / dMP , which the stock-margin estimates bound: under realized-rate scaling they reject a synchronized withdrawal of one percent at every observed quantile, so the stock-quantity channel added at most about 9.5 percent to the modeled response during our sample. That bound is a property of our standard errors and would tighten with a longer or wider panel, which is the point of reporting it as a bound: a paper that reported only the product of the two factors would appear to have a smaller result for having better data, while the slope does not move.

At $\kappa = 0.01$ the cut changes unused capacity by -1.1 percent on average and by -8.3 percent among cardholders at or above 0.80 utilization. The financing-constraint component changes from \$ - 4.70 to \$ - 8.49 per household-quarter, an increase in magnitude of 80.5 percent. The total response changes from \$ - 39.87 to \$ - 43.66, an increase in magnitude of 9.5 percent. A contraction of five percent raises the total by 48.3 percent. The last column of Table 7 expresses each increment as a fraction of the increment under full withdrawal, which shares the pre-shock state $c_{i,0}$ with every other row; a one-percent cut is 0.8 percent of the way up that ladder.

6.2 Interpreting the magnitude

The two figures just reported—80.5 percent on the financing-constraint component, 9.5 percent on the total—differ by a factor of about eight and a half, and neither is meaningful without the other. Table 8 sets out the ladder that connects them, in the manner of La Cava et al. (2016): each rung is the same experiment measured against a different denominator, and the dilution steps are visible rather than implicit.

Three readings follow. First, the population-average increment of \$3.79 per household-quarter is small for an arithmetic reason, not an economic one. The one-percent cut lowers spending for 5.7 percent of households, by a mean of \$67 each; those two numbers approximately reproduce the increment. The identity is exact within each pairing; the reported share, conditional mean, and increment are averaged separately across pairings. Nothing at all happens to the remaining households, and averaging across them is what produces the small figure. Second, the effect is concentrated where the framework says it should be: 18.6 percent of cardholders at or above 0.80 utilization are affected, against 6.9 percent of cardholders overall, and the ninetieth percentile of the additional shortfall is \$171 against a median of \$33. Third, on the broadest denominator the modeled response moves from -0.218 to -0.239 percent of mean

Table 8: From a one-percent line cut to the population average

	Lines stable	Lines cut 1%	Change
1. Authorized line A , synchronized cut	—	−1%	
2. Unused capacity K , all cardholders	—	−1.1%	
3. Unused capacity K , cardholders at $u \geq 0.80$	—	−8.3%	
4. Households whose spending falls	—	5.7%	
5. High-utilization cardholders whose spending falls	—	18.6%	
6. Mean additional shortfall among affected households	—	\$67	
7. median / 90th percentile	—	\$33 / \$171	
8. Population-average spending change	—	−\$3.79	
9. Financing-constraint component of the response	−\$4.70	−\$8.49	+80.5%
10. Total short-run response	−\$39.87	−\$43.66	+9.5%
11. Total response, share of mean baseline spending $E[c_{i,0}]$	−0.218%	−0.239%	−0.021 pp

Notes: The counterfactual of Table 7 at $\kappa = 0.01$, decomposed. Rows 1–3 are the mechanical mapping from the authorized line to usable capacity; u is credit-card utilization. Rows 4–8 are the extensive and intensive margins of the spending response: a household is *affected* when its post-shock spending is strictly lower under the cut, and its additional shortfall is the difference. Row 5 uses high-utilization cardholders as its denominator. Within each pairing, row 8 equals minus the share affected times the conditional mean shortfall. Averaging the product is not exactly the product of the separately averaged rows 4 and 6. The population mean includes zero effects for the 94.3% of households whose spending is unchanged. Rows 9–11 restate the experiment on the three denominators the paper uses elsewhere. Dollar quantities and incidence statistics are means over 50 independent cross-survey pairings at $m = 0.25$; rows 9–11 take ratios of those means. Monte Carlo standard deviations are reported in the replication package.

model-implied baseline quarterly spending of \$18,288—the weighted mean of $c_{i,0}$ in Equation (5), which is below the mean financing requirement $E[R_i]$ because constrained households do not finance R_i in full. That level is the same order as the 0.1 to 0.2 percent La Cava et al. (2016) obtain for a 100 basis-point change in Australia through the household cash-flow channel. The two exercises differ in country, in coverage—theirs is a cash-flow channel only, ours includes an ordinary income channel but is partial equilibrium—and in method, so we read the agreement as an order-of-magnitude concordance rather than a replication.

For scale on the other side of the market, Chava et al. (2023) provide the closest causal counterpart to this counterfactual: a shock that does withdraw already-authorized capacity. They estimate a marginal propensity to borrow of 53 cents out of a dollar of funding-induced credit-limit reduction for credit-constrained borrowers, close to the 45 cents Gross et al. (2020) obtain for low-credit-score borrowers around bankruptcy-flag removal. Their object is borrowing and ours is a financing shortfall, so the two are not interchangeable; what the comparison establishes is that a proportional withdrawal of authorized capacity of the size we consider is small relative to the shocks under which issuers are known to cut lines.

Pre-authorized capacity is therefore economically potent but, at the margins we observe, inert to ordinary tightening. Even small synchronized cuts in authorized lines would materially increase modeled financing shortfalls for the households nearest their limits, and Facts 2–5 say that tightening does not deliver them.

6.3 What moves the bound?

The 9.5 percent figure is conditional on the baseline calibration. Table 9 varies the assumptions behind it one at a time and reports the amplification of the same one-percent line cut; the full grids are in Appendix B. Three assumptions move the bound by several percentage points: the definition of the short-run financing requirement, the ordinary marginal propensity to consume, and whether higher rates discourage drawing. The cross-survey dependence—the one dependence the construction imposes without data—moves it by about two points, and statistical uncertainty by less than half a point.

Table 9: What moves the stock-channel bound

Assumption varied	Baseline	Alternative	Amplification of a 1% cut	Table
Baseline calibration			9.5%	7
Financing requirement R_i	Total CEX expenditure	Excluding pension contributions and cash gifts	7.9%	B.7
		Excluding housing	5.1%	
		Nondurables and services, net of vehicles	1.3%	
Ordinary MPC m	$m = 0.25$	$m = 0.15$	14.3%	B.1
		$m = 0.40$	6.3%	
		Liquid-wealth schedule	9.4%	
APR-sensitive drawing	None (hard capacity)	Gross and Souleles (2002) long-run anchor	4.6%	B.8
		Most extreme calibration	3.1%	
Cross-survey dependence $\rho_{R,K}$	Conditional random pairing	Target $\rho_{R,K}$ from -0.5 to $+0.5$	7.2–9.3%	B.3
Assignment of SCF states	Joint donor	Joint donor for (A, B) only	9.4%	B.6
		Score-based assignment	8.9%	
Statistical uncertainty		Pairing, imputation, CEX sampling	s.e. 0.45 pp	B.11

Notes: Amplification is the proportional increase in the modeled short-run spending response to a 100 basis-point tightening, $|\Delta C^{1\%}|/|\Delta C^{\text{stable}}| - 1$, when open-account authorized lines are also cut by one percent through Equation (8); every input not named in a row is at its baseline value. The APR-sensitive rows recompute the amplification inside the soft-cap economy of Section 6.5, with the cap tightening d held fixed. The dependence row varies the within-income-group rank correlation between the financing requirement and unused capacity. The score-based assignment uses an approximate common price conversion. The statistical row combines three standard errors in quadrature. Table 5 gives the direction of every other input.

Financing requirement. The baseline measures R_i with total CEX expenditure. This is broader than the nondurable concepts used in much of the MPC literature, but it reflects that housing, insurance, and pension obligations absorb current cash flow even when they are not the most adjustable consumption categories. Narrowing the concept mechanically reduces the

share of households for whom the constraint binds, and the amplification falls from 9.5 percent on total expenditure to 7.9 percent excluding pension contributions and cash gifts, 5.1 percent excluding housing, and 1.3 percent on nondurables and services net of vehicle purchases. The amplification tracks the level of total attenuation rather than being invariant to it, so the headline figure is a statement about the broad requirement concept; we report it that way and do not select the definition to preserve it. Unused capacity still materially relaxes financing constraints under every concept: the attenuation of the constraint component moves only from 39.2 to 32.6 percent across them (Appendix B.4).

Ordinary MPC. The amplification divides a constraint-side increment by the *total* response, which is 88.2 percent income channel at the baseline, so a larger ordinary MPC dilutes the same increment. It runs from 14.3 percent at $m = 0.15$ to 6.3 percent at $m = 0.40$, and is 9.4 percent under a liquid-wealth schedule. The attenuation of the constraint component in the no-capacity comparison has the opposite sensitivity: it compares two constraint components, both of which shrink with m , and holds at 39.19 percent as m ranges from 0.10 to 0.40. The range of the amplification across m is wider than its range across the cross-survey dependence, so m is the larger source of uncertainty for the stock-channel bound even though it is the smaller one for the attenuation (Appendix B.1).

Cross-survey dependence. No survey observes a household’s financing requirement together with its authorized capacity, so conditional random pairing imposes that dependence. Moving the target within-income-group rank correlation between R and unused capacity from -0.5 to $+0.5$ keeps the amplification between 7.2 and 9.3 percent, because the same household states enter its numerator and denominator (Appendix B.2). Dependence observed *within* the SCF is preserved by the joint donor, so it is not a source of uncertainty here; assignments that do not preserve it give 9.4 and 8.9 percent. The same cross-survey dependence moves the level of the no-capacity comparison of Section 6.4 much further.

Statistical uncertainty. Pairing error, SCF imputation, and CEX sampling together give a standard error of 0.45 percentage points on the amplification (Table B.11), well inside every assumption range above.

The stock response itself. The baseline normalizes the impact response of open-account lines to zero. Passing the estimated point response and its confidence bounds through the same block instead leaves the liquidity-buffer effect economically meaningful throughout the confidence interval, with the attenuation of the constraint component between 20.9 and 90.3 percent (Table B.9). The counterfactual is also uniform across the level of the line, whereas the high-frequency estimates of Fact 2 leave open a rotation that trims only the upper quantiles. That possibility cuts two ways. For the counterfactual it can be signed: a withdrawal arriving

as a rotation would bind fewer constrained households than the uniform cut of Equation (8), so the amplification we report is the generous case rather than the conservative one. For the stable-stock baseline it is benign only on a premise we state rather than test—that the largest lines belong to households for whom capacity is slack, as they tend to in the cross-section—because a household can carry a large line and still revolve against most of it. Fact 5 bounds the incidence of forgone line increases but not their size, so that margin stays outside the block. Appendix B.6 discusses all three.

6.4 A no-capacity benchmark

The earlier literature invites a different comparison, against an economy that has no card capacity in *either* period. It measures the value of holding a line at all rather than the stability of lines, and it is secondary here for two reasons. It is not a policy derivative: the two arms differ in level as well as in response, with pre-shock spending of \$18,288 against \$17,820. And it is not point-identified.

Its economic content is nonetheless clear. In the no-capacity economy 17.9 percent of households cannot finance their observed quarterly expenditure level, against 12.6 percent in the observed economy (Figure 1), and the observed economy’s constraint-driven response is 39.2 percent *smaller*. Because capacity operates only on that component and the ordinary income channel is common to both economies, the total modeled response is 7.1 percent smaller (Figure 2). Within each pairing that total is the no-card constraint share of the response, 18.0 percent, times the constraint attenuation; the second factor is nearly unchanged in m and in δ_y , so what moves the total is how much of the response financing constraints generate. The total uses the dollar convention; as a share of each economy’s own baseline spending it is 9.4 percent, so the convention we use understates the comparison rather than flatters it. Appendix B.1 reports the decomposition across MPC calibrations.

Neither share is a calibration target. The framework matches no moment of the constrained population, which is what makes the implied shares usable as an external check on the financing rule. The relevant external object is the poor hand-to-mouth share of Kaplan et al. (2014)—households holding neither liquid nor illiquid wealth—rather than their total hand-to-mouth share, because their wealthy hand-to-mouth hold illiquid assets they can convert at a cost. Pooling the 1989–2010 SCF, they put that share at 12.1 percent under a credit limit imputed at one month of income, and at 11.4 percent when the limit is instead each household’s self-reported SCF credit limit, the construction ours uses. Our 12.6 percent is the comparable figure, since both count unused card capacity as spendable; the no-capacity share is not. The definitions differ—theirs is a threshold on liquid wealth relative to pay frequency, ours asks whether an observed quarterly expenditure level can be financed—so the closeness of the two numbers is partly fortuitous. What the comparison rules out is the failure that would matter here. The amplification in Section 6.1 is generated entirely by households at the kink of Equation (5), so a

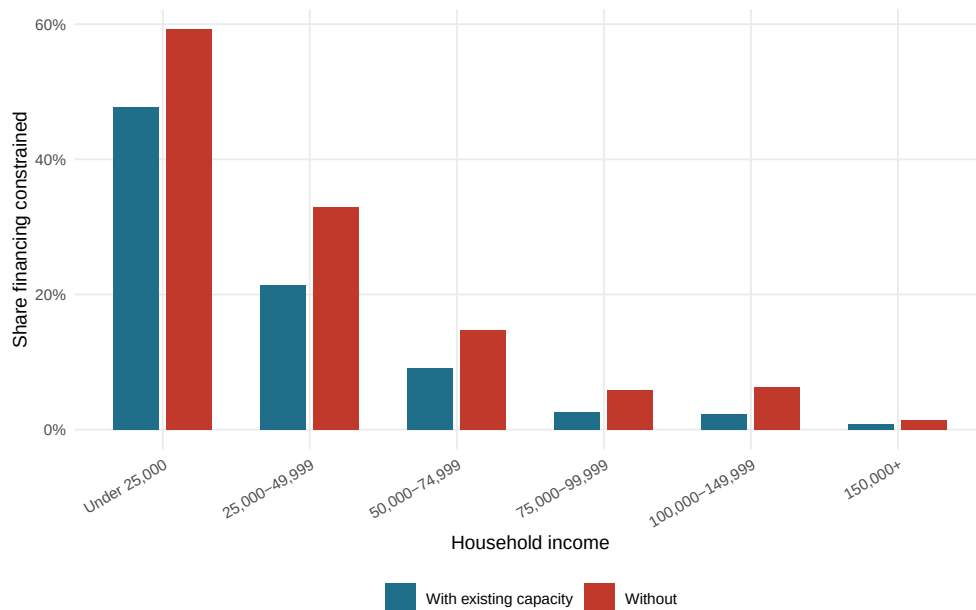


Figure 1: Financing constraints with and without unused card capacity

Notes: Weighted shares in one representative donor pairing, as in Table 6; headline aggregates average over 50 pairings.

framework that placed several times more or fewer households there than the external evidence admits would over- or understate that amplification for a purely mechanical reason. It does not.

Pre-authorized capacity is most valuable on impact, because it is already in place: a household without a line can eventually substitute toward liquid savings, but doing so takes time. Appendix B.7 reports a rebuilding scenario disciplined by CEX saving capacity. One feature of it does not depend on the timing assumption: measured saving rates are negative in the two lowest income groups, so the households most likely to be financing constrained are also those least able to replace a lost line through saving.

What the benchmark cannot deliver is a point estimate. Its level turns on the dependence between a household's financing requirement and its authorized capacity, which no dataset observes: moving the target rank correlation between R and K from -0.5 to $+0.5$ moves the attenuation of the constraint component from 23.4 to 61.8 percent (Table B.2), and narrowing the financing requirement lowers total attenuation from 7.1 to 0.8 percent (Appendix B.4). Unused capacity materially relaxes financing constraints under every construction; how much it attenuates the total response is a range, and Section 6.5 shows that APR-sensitive drawing can reverse its sign.

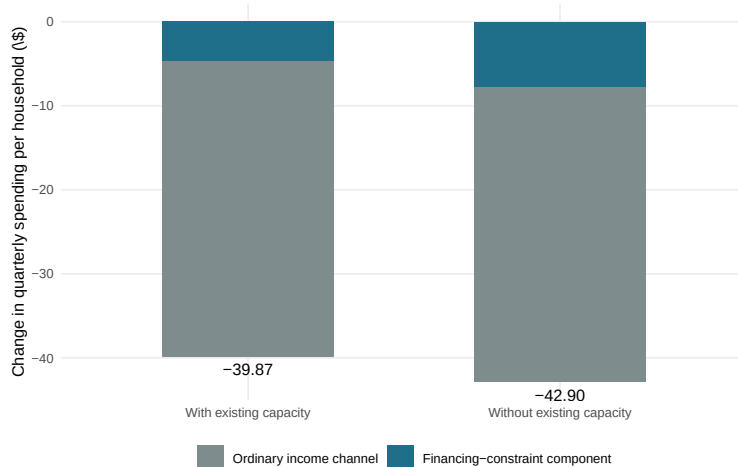


Figure 2: Ordinary and financing-constraint components of the consumption response

Notes: Quarterly dollars per household at $m = 0.25$ and $\delta_y = 0.54$ percent. Labels give the total response in each economy.

6.5 Borrowing before the line binds

The baseline treats authorized capacity as available up to the line. A higher APR may reduce willingness to borrow before that point, and this is the most consequential assumption in the quantitative exercise. We replace the hard cap with a behavioral borrowing cap, the reduced form of a convex borrowing cost set out in Appendix B.5, and calibrate it to external evidence rather than estimating it. In terms of Equation (4), the exercise imposes an external value on the household response to the price term, whose policy response Fact 1 does estimate, and asks whether the sensitivity of spending to an additional stock contraction survives once the observed APR increase also reduces desired drawing.

The cap has a level and a slope, and they pose different questions. The level asks whether treating the line as hard capacity overstates the buffer. Holding a cap fixed across the shock, even one at 70.3 percent of unused capacity leaves the attenuation of the constraint component at 33.4 percent against 39.2 percent at the line, so the level of the result does not depend on capacity being a hard cap. The slope lets the estimated APR increase itself tighten the cap by d dollars after the shock, and this is the margin the interest-elasticity literature disciplines. Gross and Souleles (2002) imply that a one percentage-point increase in the account interest rate lowers the debt-to-limit ratio by about 1.7 percentage points in the long run; scaled to the 1.11 percentage-point APR increase of Fact 1, that is a decline in card borrowing of 8.7 percent, against 5.8 percent at their two-month horizon.

At the long-run anchor the two statistics of this section part company. The offsetting channel is -40.94 dollars per household per quarter. That more than eliminates the buffer in the no-capacity comparison: ΔC becomes $\$-80.81$ against $\$-42.90$ without cards, so total attenuation

is -88.3 percent and the modeled response is *larger* with card capacity than without it. Measured as the uniform authorized-line cut that would deliver the same aggregate response, the long-run anchor is equivalent to a cut of 10.6 percent, an order of magnitude above the 0.8 percent cut that reproduces the no-card total response and far outside the estimated open-account interval. The stock-channel bound does not reverse. Recomputed inside the soft-cap economy with d held fixed, the amplification of a one-percent authorized-line cut falls from 9.5 percent at $d = 0$ to 4.6 percent at the long-run anchor and 3.1 percent at the most extreme calibration, and it changes sign nowhere on the grid (Table B.8). The borrowing-cost channel enters both arms of the line-cut experiment and only one arm of the no-capacity comparison: cutting an authorized line withdraws resources from households whose spending is constrained, whether or not they were already discouraged from drawing. Withdrawing authorized lines would still have amplified the response, at roughly half the magnitude.

Two considerations bound how much weight the calibration deserves. The elasticity we borrow is a nine- to twelve-month response of the *stock* of revolving debt, whereas the object here is one quarter's flow of constraint-relieving borrowing, so mapping one onto the other one for one is an upper bound on the channel rather than a point estimate of it. And a behavioral cap and an issuer line cut need not bind the same households. Here the two incidences are close without coinciding: among households constrained under at least one of the two experiments, 96.9 percent are constrained under both (Appendix B.5). Conditional on the willingness to draw on an existing line being unchanged by the policy rate, a withdrawal of authorized capacity would have removed a measurable amount of short-run liquidity. Whether the net effect of card capacity on transmission is positive depends on a borrowing-cost elasticity we do not estimate, and Section 7 states what would identify it.

7 Conclusion

Revolving credit has a price, a flow, and a stock, and monetary tightening changes them differently. Card borrowing reprices rapidly. Authorized capacity on open accounts does not contract at any of the four margins through which it could: the estimates show no downward shift in the limit distribution and reject contractions as small as one percent at every observed quantile, line cuts do not become more frequent, accounts do not leave the stock faster, and line increases are not suppressed at the scales we can test. A bank holding-company panel spanning four tightening cycles gives the same answer for aggregate committed capacity. Median initial limits on newly opened accounts do fall relative to open-account limits on impact under high-frequency identification, with no shift in the credit scores of new borrowers, and households make greater use of the capacity they already hold. Estimating tightening and easing separately leaves this picture intact: a rate cut lowers the APR by about as much as a hike raises it, and moves authorized limits no more than a hike does.

This asymmetry—between existing and new borrowers, not between the two directions of policy—changes the interpretation of the consumer-credit channel. Tightening reaches revolving credit through repricing and through the terms offered at origination, and its quantity effect appears at origination rather than through the withdrawal of commitments already made. Marginal credit supply can therefore tighten without withdrawing the liquidity households already have. The accounting exercise prices the stock-quantity channel this leaves inactive. Under the baseline calibration, and holding fixed households’ willingness to draw on an existing line, a withdrawal as large as the one percent our estimates rule out would have raised the modeled short-run response by at most 9.5 percent, with the loss concentrated among high-utilization cardholders whose unused capacity is thinnest. That bound depends more on how broadly the short-run financing requirement is defined and on the ordinary marginal propensity to consume than on the cross-survey match, and it roughly halves, without changing sign, once higher rates are allowed to discourage drawing.

The broader lesson is not that revolving credit weakens monetary policy in all states. Lines are revocable, and other shocks—notably disruptions to lender funding—do induce issuers to cut them (Chava et al., 2023). Ordinary tightening need not operate through the same margin as a financial crisis. What we establish is that the stock of authorized capacity survives ordinary tightening at the margins the data let us observe, and that the stock is economically potent as liquidity: were it withdrawn, modeled financing shortfalls would rise materially. Ordinary tightening does not withdraw it by enough for that stock-quantity channel to be large.

Two questions follow directly, and both are answerable with data that exist. The first is whether the household-level spending response varies with pre-shock unused capacity. Our evidence is distributional, and establishing household-level heterogeneity requires account- or borrower-level panel data that jointly observe authorized capacity and subsequent spending or borrowing. The second is more consequential for the quantitative claim. Section 6.5 shows that if a higher APR discourages drawing on an existing line at the elasticities estimated for revolving balances, the offsetting channel is large enough to reverse the sign of the net effect of card capacity on transmission. What would settle it is the response of *drawn balances* on existing accounts around policy dates, at the account level, holding the authorized line fixed—and the FR Y-14M collection observes exactly that pair. The two questions share one data requirement, and answering them is the natural next step rather than a refinement.

The contribution here is narrower and, we think, robust to how those questions resolve. In revolving credit, the quantity channel of monetary policy is concentrated at origination rather than in the withdrawal of previously authorized commitments. Because pre-authorized capacity remains available, marginal credit can tighten without removing an important source of short-run household liquidity.

A Data and Empirical Supplements

A.1 Data sources and coverage

Table A.1: Data sources and coverage

Product	Source	Coverage used
Large Bank Credit Card and Mortgage Data	Federal Reserve Bank of Philadelphia; built from FR Y-14M schedule D	2012Q3–2025Q4, quarterly
Card-market panel merged with policy surprises	Derived: card panel + FRED macro controls + surprise series	2012Q3–2025Q4, quarterly
Quarterly monetary-policy surprises	SF Fed USMPD; Bauer and Swanson (2023); Jarociński and Karadi (2020); Miranda-Agrippino–Ricco; Wu–Xia	1990Q1–2025Q4, quarterly
FR Y-9C card commitments	Chicago Fed/NIC BHC filings; BHCK3815 and BHCKJ455	1990Q3–2018Q2, quarterly
Effective federal funds rate	FRED series FEDFUNDS	monthly, averaged to quarters
Consumer Price Index, all items	FRED series CPIAUCSL	monthly, averaged to quarters
Savings deposit rate	FRED series SAVNRNJ; ends 2021Q1	monthly, averaged to quarters
Diary of Consumer Payment Choice, 2024	Federal Reserve Bank of Atlanta, individual-level public file	2024 diary wave, household file
Survey of Consumer Finances, 2016	Board of Governors, full public file (scf2016s) + Summary Extract (SCFP2016)	2016 wave, reflatd to 2024 dollars
Survey of Consumer Finances, 2019	Board of Governors, full public file (scf2019s) + Summary Extract (SCFP2019)	2019 wave, reflatd to 2024 dollars
Survey of Consumer Finances, 2022	Board of Governors, full public file (scf2022s) + Summary Extract (SCFP2022)	2022 wave, reflatd to 2024 dollars
Consumer Expenditure Survey, Interview PUMD	BLS, intrvw23 package	interview years 2023–2024

Notes: The card-market product is the Philadelphia Fed’s Large Bank Credit Card and Mortgage Data, which aggregates FR Y-14M reports from the largest card-issuing institutions subject to that collection’s reporting threshold; it is therefore a large-bank aggregate and not the universe of the general-purpose card market, and it carries no account identifier. The Survey of Consumer Finances enters as three pooled waves; we use it to match marginal and conditional distributions of liquid assets, authorized lines, and utilization rather than to estimate SCF regressions. The preferred household construction carries liquid assets, authorized limits, and balances jointly from one weighted SCF donor within the matching cell, preserving the within-SCF dependence among the balance-sheet states. The donor pool pools all five implicates with the public weight divided by five, so the point estimates average over the Board’s imputation rather than conditioning on one draw; Table B.11 reports the between-implicate spread around that average.

Price base. The preferred household block is assembled from three surveys whose dollar values sit on different bases, and we put all of them on a common 2024 base, the year of the Diary anchor. The authorized line, the balance, and the income that defines the matching cell come from the SCF full public files and are survey-year nominal; liquid assets come from the Summary Extract, which the Board has already expressed in the dollars of the most recent survey using one constant factor per wave (1.2493, 1.1802 and 1.0809 for 2016, 2019 and 2022). We recover that factor from the data and assert it, carry the nominal quantities to 2022 dollars with it, and then carry the whole SCF block from 2022 to 2024 with CPIAUCSL; the CEX is deflated from its interview quarter to the 2024 annual average. Income cells are cut on the reflatd income, not on the nominal income. Card-market limits, which enter only the regressions of Section 3, are deflated by CPIAUCSL throughout and are unaffected. Table B.13 compares the preferred common-price baseline with the mixed-base convention. Of the Diary of Consumer Payment Choice we use only the individual-level file, which supplies the household weight, income group, card access, and revolving status; the transaction-level file is not an input to this paper. The Diary is frozen at the 2024 wave. The Consumer Expenditure Survey is the `intrvw23` package, whose interview records span calendar 2023 and 2024.

The card-market panel enters the local projections of Section 3 with 43 of the 54 quarters usable after differencing and lagging in the APR regressions and 42 in the limit regressions, whose series begins one quarter later. Against the seven regressors of the benchmark specification the latter leave 35 degrees of freedom, which determine the Student- t critical values for those limit regressions; other regressions use their own residual degrees of freedom. Newey–West standard errors use a lag truncation of $\min(h + 1, 8)$ at every horizon, so the impact-horizon estimates that carry the headline results use one lag. The FR Y-9C analysis sample covers 70,523 institution-quarters for 1,911 institutions across 112 calendar quarters; differencing and lagging leave up to 110 quarters usable in any single regression. The commitment item is reported as BHCK3815 before 2010 and BHCKJ455 afterward. The two definitions differ materially, with a within-institution median ratio of 0.823 at the break, so we drop differences that span it rather than splice the series.

A.2 Equivalence bounds by identification

Table A.2: Equivalence bounds on open-account capacity, by identification

Margin	Estimate (%)	95% CI	p -value, $H_0 : \beta \leq -1\%$
<i>Realized Δ fed funds</i> (per 100 basis points)			
Open-account $p50$	0.62	$[-0.50, 1.75]$	0.003
Open-account $p75$	-0.15	$[-0.86, 0.56]$	0.010
Open-account $p90$	-0.27	$[-0.93, 0.40]$	0.015
<i>SF Fed USMPD, market events</i> (one s.d. = 6.3 basis points)			
Open-account $p50$	4.83	$[-3.13, 13.45]$	0.076
Open-account $p75$	-0.23	$[-6.77, 6.78]$	0.409
Open-account $p90$	-2.70	$[-6.90, 1.70]$	0.783
<i>Bauer–Swanson, orthogonalized</i> (one s.d. = 7.9 basis points)			
Open-account $p50$	9.26	$[1.62, 17.48]$	0.005
Open-account $p75$	0.71	$[-2.67, 4.21]$	0.158
Open-account $p90$	-2.15	$[-4.22, -0.03]$	0.862
<i>Jarociński–Karadi, pure MP</i> (one s.d. = 8.5 basis points)			
Open-account $p50$	3.15	$[-7.28, 14.75]$	0.220
Open-account $p75$	-8.49	$[-13.95, -2.68]$	0.993
Open-account $p90$	-3.34	$[-8.88, 2.52]$	0.792
<i>SF Fed USMPD, statement</i> (one s.d. = 5.5 basis points)			
Open-account $p50$	4.71	$[-6.82, 17.67]$	0.169
Open-account $p75$	-2.42	$[-11.49, 7.57]$	0.617
Open-account $p90$	-6.56	$[-11.85, -0.96]$	0.974

Notes: Log-response coefficients and their standard errors are rescaled to a 100 basis-point policy change by multiplying their one-standard-deviation values by 100/s.d., with the s.d. in basis points. Estimates and interval endpoints are then reported as $100(e^\beta - 1)$. Rescaling leaves the test of $\beta = 0$ unchanged but widens the interval relative to the fixed -1 percent threshold. The high-frequency surprises have standard deviations of 5.5 to 8.5 basis points, so a 100 basis-point equivalence statement requires a twelve- to eighteen-fold extrapolation. Of the 12 high-frequency open-account cells, 1 rejects a one-percent contraction at the five percent level, against three of three under realized changes. Three cells are significantly negative on impact: Jarociński–Karadi at $p75$, and Bauer–Swanson and the statement window at $p90$. The sign pattern is monotone in the quantile—no series is negative at $p50$, three of four are at $p75$, and all four are at $p90$ —so under high-frequency identification the open-account limit distribution rotates rather than shifts. Section 3 discusses what that does and does not establish. The Jarociński–Karadi cell is robust to the control set and is not distinguishable from zero at any later horizon. Specification as in Table 2; Newey–West standard errors with lag $\min(h + 1, 8)$, so lag 1 at the impact horizon reported here; Student- t critical values.

A.3 Sample splits

Table A.3: Sample splits and a full-sample regime interaction

Sample	APR	Open $p50$	New $p50$	Difference	n
<i>Panel A. Period subsamples, estimated separately</i>					
Full 2012–2025	1.11 (0.00)	0.62 (0.27)	−2.81 (0.33)	−3.43 (0.20)	43/42
2012–2019	0.85 (0.01)	4.49 (0.05)	26.65 (0.00)	22.16 (0.01)	19/19
Excl. 2020–21	1.05 (0.00)	0.76 (0.44)	3.90 (0.44)	3.14 (0.52)	34/33
2022–2025	1.17 (0.00)	−3.55 (0.01)	−0.38 (0.97)	3.16 (0.75)	15/14
<i>Panel B. One full-sample regression, $\Delta i_t \times \mathbf{1}\{t \geq 2022\}$</i>					
Response, pre-2022 (β)	0.71	1.48	1.97	0.48	43/42
p -value	(0.001)	(0.363)	(0.820)	(0.955)	
Interaction (γ)	0.78	−1.27	−6.39	−5.12	43/42
p -value	(0.011)	(0.528)	(0.590)	(0.665)	
Response, 2022–2025 ($\beta + \gamma$)	1.49	0.21	−4.42	−4.63	43/42
p -value	(0.000)	(0.802)	(0.411)	(0.392)	

Notes: Student- t p -values in parentheses. First differences are taken only between calendar-adjacent quarters, so a subsample that omits interior years does not differ across the gap. The n column reports the APR and limit regression counts, which differ because the limit series starts one quarter later. Panel A estimates the full seven-parameter specification separately within each window; the two period subsamples have between 14 and 19 usable observations against seven parameters. Panel B estimates one regression on all 42 usable quarters, adding $\Delta i_t \times \mathbf{1}\{t \geq 2022\}$ and a post-2021 level shift while holding the control coefficients common across regimes; γ is the test of equal policy responses in the two regimes, which Panel A cannot provide. The open-account interaction is −1.27 with $p = 0.528$: the apparent reversal between Panel A’s two windows is not distinguishable from sampling variation once the specification is not refitted inside each window. Table A.7 Panel B reports the same 2022–2025 open-account estimate across control sets.

A.4 Margins of incumbent-account management

Table A.4 collects the three margins through which an open account could lose capacity without its limit distribution shifting, under all five policy measures, and adds the asymmetric realized-rate specification for the line-increase share. Table A.5 reports the economic bounds on suppressed line increases and Table A.6 the horizon path.

Table A.4: Margins through which open accounts could lose capacity

Outcome	Realized Δ fed funds	SF Fed market events	Bauer– Swanson	Jarociński– Karadi	SF Fed statement
Line-increase share (pp)	0.376 (0.175)	−0.019 (0.054)	−0.058 (0.075)	0.001 (0.089)	0.004 (0.066)
Line-decrease share (pp)	−0.004 (0.047)	0.034 (0.030)	0.034 (0.033)	−0.012 (0.066)	0.024 (0.041)
Net exit rate (pp)	0.028 (0.424)	−0.323 (0.195)	−0.279 (0.180)	−0.343 (0.431)	−0.463 (0.228)
<i>Asymmetric specification, line-increase share</i>					
Tightening component T_t	−0.004 (0.190)			—	
Easing component E_t	−0.950 (0.337)			—	
Easing component E_t , excluding 2020	−0.140 (0.123)			—	

Notes: Impact responses in percentage points. Column 1 is per 100 basis points of realized federal-funds change; columns 2–5 are per one-standard-deviation monetary-policy surprise and are not rescaled to 100 basis points. The net exit rate is Equation (2); its column-1 entry reproduces the estimate quoted in Section 3. The asymmetric rows estimate $\Delta I_t = \alpha + \beta_T T_t + \beta_E E_t + \Gamma X_{t-1} + \varepsilon_t$ with $T_t = \max\{\Delta i_t, 0\}$ and $E_t = \max\{-\Delta i_t, 0\}$, on realized changes only, with 26 tightening and 14 easing quarters; a symmetric linear response implies $\beta_T + \beta_E = 0$, which is rejected at $p = 0.037$ on the full sample. The last row shows that the easing coefficient is a pandemic artifact: dropping calendar 2020 moves it from −0.95 to −0.14 and it is no longer distinguishable from zero. Controls, trend, and Newey–West inference follow Table 1.

Table A.5: Economic bounds on suppressed line increases

Threshold c (pp)	As a share of mean incidence (%)	p -value, $H_0 : \beta \leq -c$	Rejected at 5%?
<i>Panel A. Full sample ($n = 43$)</i>			
0.25	8.2	<0.001	yes
0.50	16.4	<0.001	yes
1.00	32.8	<0.001	yes
0.30	10.0	<0.001	yes
0.76	25.0	<0.001	yes
1.52	50.0	<0.001	yes
<i>Panel B. Excluding calendar 2020 ($n = 39$)</i>			
0.25	8.2	0.002	yes
0.50	16.4	<0.001	yes
1.00	32.8	<0.001	yes
0.30	10.0	<0.001	yes
0.76	25.0	<0.001	yes
1.52	50.0	<0.001	yes

Notes: One-sided tests of the null that monetary tightening reduces the share of open accounts receiving a credit-line increase by at least c percentage points per 100 basis points. Small p -values reject suppression of that size. The first three rows use the fixed thresholds; the last three set c at 10, 25, and 50 percent of the mean quarterly incidence, which is 3.04 percentage points. Specification, controls, sample convention, and Newey–West inference are identical to Table 1; Student- t critical values with the regression’s residual degrees of freedom. Panel B drops calendar 2020, when the line-increase share fell by two thirds alongside the fastest easing in the sample.

Table A.6: Line-increase share at horizons zero through four

Policy measure	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$
Realized Δ fed funds	0.376 (0.175)	0.440 (0.314)	0.322 (0.402)	0.255 (0.330)	0.045 (0.232)
SF Fed, market events	-0.019 (0.054)	0.116 (0.140)	0.064 (0.124)	-0.062 (0.097)	0.011 (0.073)
Bauer–Swanson	-0.058 (0.075)	0.147 (0.165)	0.123 (0.126)	0.043 (0.072)	0.109 (0.078)
Jarociński–Karadi	0.001 (0.089)	-0.107 (0.115)	-0.099 (0.102)	-0.198 (0.137)	-0.021 (0.152)
SF Fed, statement	0.004 (0.066)	0.185 (0.219)	0.119 (0.192)	-0.045 (0.115)	0.026 (0.106)

Notes: Local projections of the cumulative change in the share of open accounts receiving a credit-line increase, $I_{t+h} - I_{t-1}$, in percentage points. Row 1 is per 100 basis points of realized federal-funds change; rows 2–5 are per one-standard-deviation surprise and are not rescaled. Controls, trend, and Newey–West inference follow Table 1; the lag truncation is $\min(h + 1, 8)$, as elsewhere in the paper. Newey–West standard errors in parentheses. The main text reports $h = 0$ only: the later coefficients are not stable enough to read as a dynamic profile.

A.5 Identification of the new-minus-open-account difference

Table A.7: New minus open-account limit difference across control sets

Policy measure	Full controls	Trend only	No controls
Realized Δ fed funds (100bp)	-3.43 (0.20)	2.24 (0.45)	2.30 (0.44)
SF Fed USMPD, market events (1SD)	-2.64 (0.12)	-4.74 (0.06)	-4.80 (0.06)
Bauer–Swanson, orthogonalized (1SD)	-6.41 (<0.001)	-7.74 (<0.001)	-7.34 (<0.001)
Jarociński–Karadi, pure MP (1SD)	-1.80 (0.49)	-1.93 (0.52)	-0.81 (0.75)
SF Fed USMPD, statement (1SD)	-3.98 (0.05)	-6.25 (0.01)	-6.07 (0.02)

Panel B. Open-account p50 across control sets, by window

Control set	2022–2025	Full 2012–2025
Full controls	-3.55 (0.01)	0.62 (0.27)
Trend only	0.30 (0.75)	0.18 (0.73)
No controls	-0.13 (0.82)	0.12 (0.85)
Drop utilization	-3.54 (0.01)	0.43 (0.41)
Drop delinquency	-3.35 (0.03)	0.18 (0.73)
Macro only (u, VIX, trend)	-3.07 (0.03)	0.19 (0.73)

Notes: Panel A reports the percent response of $\Delta \log A_{p50}^{new} - \Delta \log A_{p50}^{open}$; Student- t p -values in parentheses. Full controls are lagged unemployment, VIX, delinquency, utilization, and a linear trend. First differences are taken only between calendar-adjacent quarters. The realized-rate estimate changes sign across control sets; the high-frequency estimates do not. Panel B reports the open-account $p50$ response itself, which Table A.3 estimates at -3.55 percent in 2022–2025 under the full control set. That estimate is not robust: it ranges from -3.55 to 0.30 percent across the six sets, is distinguishable from zero in four of the six, and is essentially zero under trend-only and no-controls specifications. The full-sample estimate is stable and near zero throughout. Panel B has 14 usable observations in the 2022–2025 window against up to seven parameters.

A.6 Composition of new accounts

Table A.8: Composition of newly originated accounts after a monetary-policy surprise

Outcome	SF Fed market events	Bauer– Swanson	Jarociński– Karadi	SF Fed statement
<i>Panel A. Borrowers at origination</i>				
Credit score, new accounts, p_{50} (points)	−0.10 (0.32)	−0.29 (0.26)	−0.53 (0.65)	−0.08 (0.30)
Credit score, new accounts, p_{25} (points)	−0.37 (0.45)	−0.42 (0.30)	−0.53 (0.79)	−0.41 (0.45)
Credit score, new accounts, p_{10} (points)	0.06 (0.46)	−0.06 (0.42)	−0.01 (1.17)	0.14 (0.58)
Credit score, open accounts, p_{50} (points)	0.07 (0.20)	−0.02 (0.13)	0.05 (0.29)	−0.05 (0.15)
Share of new accounts below 660 (pp)	0.16 (0.21)	0.16 (0.14)	0.18 (0.34)	0.20 (0.20)
Share of new commitments below 660 (pp)	0.06 (0.06)	0.06 (0.05)	0.08 (0.11)	0.06 (0.06)
New accounts opened (%)	−0.08 (1.47)	1.44 (1.25)	3.17 (2.89)	2.35 (1.27)
<i>Panel B. Median initial limit on new accounts (%)</i>				
All new accounts	−2.34 (1.67)	−5.71 (0.91)	−1.53 (2.65)	−3.73 (1.93)
Score 660–719	−4.15 (1.57)	−4.47 (1.06)	−5.16 (2.15)	−5.21 (1.39)
Score 720 and above	−2.14 (1.19)	−3.08 (1.01)	−1.20 (2.52)	−2.50 (1.21)
Score below 660	−2.08 (2.64)	−4.10 (2.49)	2.53 (2.45)	−5.75 (2.65)
<i>Panel C. New – open p_{50} (%)</i>				
As in Table 3	−2.64 (1.68)	−6.41 (0.92)	−1.80 (2.60)	−3.98 (1.96)
Controlling for Δ median score	−2.54 (1.65)	−5.95 (0.80)	−0.92 (2.40)	−3.87 (1.86)

Notes: Response at impact to a one-standard-deviation monetary-policy surprise; surprises are not rescaled to 100 basis points. Newey–West standard errors in parentheses. Specification, controls, sample rule and Student- t inference are those of Table 3; only the dependent variable changes, and the first row of Panel C reproduces that table. Credit scores are in points and shares in percentage points; limits and the account count are in log points times 100, and limits are CPI-deflated. Score bands are assigned at origination and observed only for approved accounts, so they inherit selection at approval. The median initial limit below 660 equals \$700 in 42 of the 54 quarters and carries little time-series information. The last row adds the contemporaneous change in the median origination score as a control; because that change could itself respond to policy, the row bounds the composition channel rather than estimating it. Measured against the market-wide open-account median rather than within band, the Bauer–Swanson responses are −5.17 for 660–719 and −3.78 for 720 and above. At the gradient of median initial limits between the two upper bands, composition alone would need a shift of 4.0 points in the median origination score to produce the Bauer–Swanson difference, taking a gap of 60 points between band midpoints; the figure is 3.0 at a gap of 45 points and 5.4 at 80.

A.7 High-frequency responses and horizon paths

Table A.9: Card-market responses to high-frequency policy surprises

Outcome	SF Fed market events	Bauer– Swanson	Jarociński– Karadi	SF Fed statement
Assessed APR (pp)	0.03 (0.09)	0.10 (0.06)	0.19 (0.16)	0.06 (0.10)
Open-account limit $p50$ (%)	0.30 (0.25)	0.70 (0.28)	0.26 (0.45)	0.25 (0.32)
New-account limit $p50$ (%)	−2.34 (1.67)	−5.71 (0.91)	−1.53 (2.65)	−3.73 (1.93)
New-account limit $p90$ (%)	−1.86 (1.18)	−3.45 (0.84)	−1.34 (2.77)	−2.90 (1.17)
Utilization $p50$ (pp)	0.23 (0.16)	0.32 (0.12)	0.39 (0.30)	0.38 (0.14)
Line-cut share (pp)	0.03 (0.03)	0.03 (0.03)	−0.01 (0.07)	0.02 (0.04)
Line-increase share (pp)	−0.02 (0.05)	−0.06 (0.08)	0.00 (0.09)	0.00 (0.07)
New − open $p50$ (%)	−2.64 (1.68)	−6.41 (0.92)	−1.80 (2.60)	−3.98 (1.96)

Notes: Response to a one-standard-deviation monetary-policy surprise; surprises are not rescaled to 100 basis points. Newey–West standard errors in parentheses. Dollar limits are CPI-deflated. Limit responses are reported in log points times 100, which keeps the difference row exactly additive; for these magnitudes it differs from $100(e^\beta - 1)$ by at most 0.2 percentage points. The final row is estimated directly on $\Delta \log A_{p50}^{new} - \Delta \log A_{p50}^{open}$ and reproduces Table 3 exactly.

The ninetieth-percentile new-account limit response is +3.67 percent per 100 basis points under realized rate changes, with a confidence interval of $[-2.46, 10.18]$, and the corresponding new-minus-open-account difference is positive and imprecise. We therefore do not use it as evidence on the flow margin.

Table A.10: New minus open-account limit difference by horizon, high-frequency surprises

Policy surprise	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$
SF Fed USMPD, market events	−2.64 (0.12)	−2.04 (0.40)	−0.56 (0.80)	0.99 (0.67)	−5.07 (0.01)
Bauer–Swanson, orthogonalized	−6.41 (<0.001)	−7.04 (<0.001)	−2.48 (0.34)	1.20 (0.55)	−4.62 (0.02)
Jarociński–Karadi, pure MP	−1.80 (0.49)	−1.13 (0.72)	−0.56 (0.90)	3.18 (0.27)	−4.58 (0.13)
SF Fed USMPD, statement	−3.98 (0.05)	−3.52 (0.17)	−0.60 (0.84)	2.95 (0.29)	−5.27 (0.13)

Notes: Response of $\Delta \log A_{p50}^{new} - \Delta \log A_{p50}^{open}$ per one-standard-deviation surprise, in log points times 100; two-sided p -values in parentheses. The dependent variable is the cumulative change from $t - 1$ to $t + h$; Newey–West standard errors with lag $\min(h + 1, 8)$ and Student- t critical values. Specification, controls, and sample rule are those of Table 3. The difference is negative under all four series at every horizon through two quarters (12 of 12 cells), and is distinguishable from zero at the impact horizon for 2 of the four series. Beyond impact, precision is uneven and the point estimates are non-monotone, so we do not interpret the later coefficients as a stable dynamic response profile.

Table A.11: Open-account limits by horizon, Jarociński–Karadi surprise

Margin	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$
Open-account $p50$	0.26 (0.56)	−0.02 (0.98)	−0.11 (0.90)	0.77 (0.45)	0.89 (0.41)
Open-account $p75$	−0.75 (0.01)	−0.51 (0.29)	−0.75 (0.14)	−0.42 (0.46)	0.01 (0.99)
Open-account $p90$	−0.29 (0.25)	−0.14 (0.60)	0.00 (1.00)	0.04 (0.90)	−0.83 (0.01)

Notes: Response per one-standard-deviation Jarociński–Karadi pure-policy surprise, in log points times 100; two-sided p -values in parentheses. Specification as in Table A.10. The seventy-fifth-percentile response is distinguishable from zero at the impact horizon ($p = 0.006$) and at no later horizon (smallest $p = 0.139$), which is the basis for describing it as an impact-only estimate.

A.8 Unused card commitments by reporting-item definition

Table A.12: Unused card commitments by reporting-item definition

Reporting item	$h = 0$		$h = 2$		$h = 4$	
	Estimate	N	Estimate	N	Estimate	N
1990–2009: BHCK3815	−0.04 (0.45)	58,554	0.22 (0.79)	53,992	0.61 (1.25)	49,688
2010–2018: BHCKJ455	−1.01 (4.40)	8,982	−5.96 (7.05)	7,974	2.33 (9.71)	7,057

Notes: Response to a realized 100 basis-point change in the federal funds rate, in percent, estimated separately within each reporting-item regime. BHCK3815 covers credit-card lines extended for household *and* commercial or industrial purposes, so the pre-2010 panel is not a pure consumer-card measure; BHCKJ455, in force from 2010Q1, is defined as unused consumer credit-card lines. The post-2010 subsample has less than one sixth as many usable impact observations; its estimates are imprecise, so neither regime on its own distinguishes the response from zero. Usable institution-quarters are reported separately for each horizon and decline with h because an observation enters only when every quarter from $t - 1$ to $t + h$ is present and calendar-adjacent. Specification otherwise as in Table 4.

A.9 Tightening and easing estimated separately

Table A.13 reports the sign-split specification of Section 3.7 for the repricing margin and the three open-account limit quantiles, together with the easing-side mirror of the equivalence bounds.

Table A.13: Tightening and easing estimated separately

Outcome	n	Hike, per 100bp		Cut, per 100bp		Symmetry	Joint
		β_T	95% CI	β_E	95% CI	p	p
<i>Panel A. Full sample</i>							
Assessed-interest APR (pp)	43	1.35	[0.95, 1.74]	-0.77	[-1.45, -0.09]	0.244	<0.001
Open-account limit, $p50$ (%)	42	0.62	[-1.48, 2.76]	-0.61	[-3.38, 2.23]	0.999	0.549
Open-account limit, $p75$ (%)	42	-0.10	[-1.59, 1.41]	0.23	[-2.04, 2.55]	0.940	0.913
Open-account limit, $p90$ (%)	42	-0.03	[-1.17, 1.12]	0.63	[-1.04, 2.32]	0.636	0.621
<i>Panel B. Excluding calendar 2020</i>							
Assessed-interest APR (pp)	39	1.29	[0.78, 1.80]	-1.20	[-2.06, -0.34]	0.870	<0.001
Open-account limit, $p50$ (%)	38	0.73	[-1.58, 3.10]	-2.44	[-4.89, 0.08]	0.390	0.057
Open-account limit, $p75$ (%)	38	0.03	[-1.79, 1.89]	-0.31	[-3.05, 2.50]	0.885	0.963
Open-account limit, $p90$ (%)	38	0.45	[-0.89, 1.81]	-0.44	[-1.83, 0.97]	0.995	0.505
<i>Panel C. One-sided bounds on the easing coefficient, full sample</i>							
Open-account limit	$c = 1\%$	$c = 2.5\%$	$c = 5\%$				
$p50$	0.127	0.017	<0.001				
$p75$	0.250	0.027	<0.001				
$p90$	0.328	0.016	<0.001				

Notes: Panels A and B split the realized quarterly federal-funds change into $T_t = \max\{\Delta i_t, 0\}$ and $E_t = \max\{-\Delta i_t, 0\}$ and estimate $\Delta y_t = \alpha + \beta_T T_t + \beta_E E_t + \Gamma X_{t-1} + \varepsilon_t$, so β_T is the response to a 100 basis-point hike and β_E the response to a 100 basis-point cut. A symmetric linear response implies $\beta_T + \beta_E = 0$. Limit responses are reported as $100(e^\beta - 1)$; the APR is already in percentage points. Controls, sample convention, first-difference calendar-adjacency rule, Newey–West standard errors at lag one and Student- t critical values are those of Table 1; collapsing T_t and E_t back to Δi_t reproduces Table 1 exactly. Sample sizes differ by one because the limit series is missing one quarter that the APR series has: the repricing regression uses 26 tightening and 14 easing quarters in Panel A (25 and 11 in Panel B), the limit regressions 26 and 13 (25 and 10). Panel C reports one-sided tests of $H_0 : \beta_E \geq \log(1 + c/100)$, the easing-side mirror of the tightening bounds behind Fact 2: small p -values reject an expansion of authorized limits of at least c percent per 100 basis-point cut. The symmetry column tests $\beta_T + \beta_E = 0$ and the joint column tests $\beta_T = \beta_E = 0$. Easing coefficients should be read as bounds rather than as point estimates; Section 3.7 explains why.

The asymmetric specification splits the realized policy change into $T_t = \max\{\Delta i_t, 0\}$ and $E_t = \max\{-\Delta i_t, 0\}$, so $\beta_T < 0$ would say that hikes suppress line increases and $\beta_E > 0$ that cuts encourage them. Neither pattern appears. The tightening coefficient is -0.004 and indistinguishable from zero; the easing coefficient is -0.95 , which has the wrong sign for a monetary interpretation and is an artifact of the pandemic. With 26 tightening and 14 easing quarters in 43 observations, and with the easing coefficient moving to -0.14 once calendar 2020 is dropped, we treat the asymmetry as uninformative rather than as evidence that easing lowers line increases.

Splitting the high-frequency surprises by sign is possible, and it is uninformative for a reason worth naming: concentration rather than count. The easing side carries most of the squared variation in each series, but that variation sits in very few quarters. For the Bauer–Swanson series a single quarter supplies 82.1 percent of it, an effective sample size of 5.75 quarters; for the SF Fed statement series the figures are 68.7 percent and 6.94. Across four surprise series and eleven outcomes, 5 of 88 sign-split coefficients reach the five percent level, against 4.40 expected by chance, and symmetry is rejected in 1 of 44 regressions. We therefore report the high-frequency

split as uninformative rather than as evidence either for or against sign asymmetry.

B Quantitative Supplements

This appendix collects the full grids behind Section 6. Table 9 summarizes what they imply for the stock-channel bound.

B.1 The decomposition across MPC calibrations

Table B.1 reports the ordinary and financing-constraint components of the no-capacity comparison of Section 6.4 for each MPC case. With $m = 0.25$, the income channel produces a spending response of \$ -35.17 per household-quarter. The financing-constraint component is \$ -4.70 in the card economy and \$ -7.73 without it. Total spending changes by \$ -39.87 with unused capacity and \$ -42.90 without, so capacity reduces the modeled short-run response by 7.1 percent.

Table B.1: Decomposition of the short-run consumption response

Ordinary MPC	Income channel	Constraint		Total response		Atten. of total	Ampli. of 1% cut
		With K	No K	With K	No K		
$m = 0$ (decomposition)	0.00	-6.28	-10.33	-6.28	-10.33	39.2%	+60.4%
$m = 0.15$	-21.10	-5.33	-8.77	-26.43	-29.87	11.5%	+14.3%
$m = 0.25$ (baseline)	-35.17	-4.70	-7.73	-39.87	-42.90	7.1%	+9.5%
$m = 0.40$	-56.27	-3.76	-6.18	-60.03	-62.45	3.9%	+6.3%
Liquid-wealth schedule	-36.10	-4.12	-6.87	-40.22	-42.97	6.4%	+9.4%

Notes: Dollars per household-quarter at $\delta_y = 0.54$ percent. The income channel is $-m_i \Delta y_i$ and is identical across the two economies by construction, so unused capacity operates only on the constraint component. The attenuation of the constraint component is 39.2 percent across the common- m rows and 40.1 percent under the liquid-wealth schedule. The last column is the amplification of the modeled total response under the one-percent authorized-line cut of Equation (8), $|\Delta C^{1\%}|/|\Delta C^{\text{stable}}| - 1$, computed on the same households and draws. It is *not* stable in m : the denominator is 88 percent income channel at the baseline, so a larger ordinary MPC dilutes the same constraint-side increment. Across $m \in [0.15, 0.40]$ the amplification runs 6.3 to 14.3 percent while the attenuation of the constraint component holds at 39.19 percent—the two statistics have opposite sensitivities to m , which is why Section 6 states them separately. The liquid-wealth schedule sets $m = 0.40, 0.25$, and 0.10 for households whose liquid assets cover under two weeks, two to twenty-six weeks, and over twenty-six weeks of income.

Within each pairing, total attenuation equals the no-card constraint share times constraint attenuation. At the displayed precision, the separately averaged factors give:

$$\underbrace{7.1\%}_{\text{attenuation of total}} \approx \underbrace{18.0\%}_{\text{constraint share of the no-card response}} \times \underbrace{39.2\%}_{\text{attenuation of the constraint component}}.$$

The second factor is nearly unchanged in m , holding at 39.19 percent as m ranges from 0.10 to 0.40 under a common marginal propensity, and reaching 40.1 percent under the liquid-wealth schedule. What changes with m is the share of total spending adjustment attributed to financing constraints. At $m = 0.15$, capacity reduces total modeled transmission by 11.5 percent; at $m = 0.40$, by 3.9 percent; under a three-group liquid-wealth schedule, by 6.4 percent. For the

attenuation, then, the MPC calibration is the smaller of the two sources of uncertainty, and Appendix B.2 shows the cross-survey dependence moves the same object considerably further.

That statement does not carry over to the amplification, and the last column of Table B.1 shows why it must be made separately. The amplification of a one-percent line cut runs from 14.3 percent at $m = 0.15$ to 6.3 percent at $m = 0.40$, and reaches 60.4 percent in the $m = 0$ decomposition. The two statistics have opposite sensitivities to m because they have different denominators. The attenuation compares two constraint components, both of which shrink with m ; the amplification divides a constraint-side increment by the *total* response, which is 88.2 percent income channel at the baseline, so a larger ordinary MPC dilutes the same increment. Neither behavior is a defect of the construction, but the amplification’s range across m is wider than its range across the cross-survey dependence (7.2 to 9.3 percent), so m is the larger source of uncertainty for the stock-channel bound even though it is the smaller one for the attenuation.

The ratio is nearly unchanged by the external income calibration. Across values of δ_y from 0.25 to 3.0 percent, a twelvefold range, the with-to-without ratio of the constraint component stays between 0.606 and 0.608; δ_y scales the dollar magnitude of the response in both economies, not their ratio.

B.2 Cross-survey dependence

The financing-constraint event $R_i > y_i + \lambda_x X_i + K_i$ is a nonlinear threshold, so it depends not only on the marginal distributions of its arguments but on their joint distribution. No survey observes a household’s financing requirement and its authorized card capacity together. The match therefore has to induce that dependence, and we report how much the answer moves when it does.

Table B.2: Attenuation under alternative cross-survey dependence structures

Target rank correlation	R paired with capacity K		R paired with liquid assets X	
	Constraint	Total	Constraint	Total
−1.00	7.4%	1.9%	31.6%	8.9%
−0.75	16.3%	3.6%	34.9%	9.1%
−0.50	23.4%	4.9%	35.7%	8.4%
−0.25	30.6%	5.9%	37.7%	7.9%
0.00	39.4%	7.1%	39.4%	7.1%
0.25	50.3%	8.3%	41.7%	6.0%
0.50	61.8%	9.4%	43.0%	4.6%
0.75	74.8%	10.2%	43.4%	2.8%
1.00	83.2%	10.7%	37.8%	1.1%

Notes: Attenuation of the financing-constraint component and of the total response at $m = 0.25$, as the within-income-group rank correlation between the financing requirement R and each balance-sheet state is varied. Donors are drawn so that the realized rank correlation approaches the target. The dependence between R and K is not observed in any single dataset; the table therefore reports how the attenuation changes under alternative dependence structures rather than treating the baseline match as point identifying the effect.

Table B.2 varies the within-income-group rank correlation between the financing requirement and each balance-sheet state. Each scenario reassigns whole CEX income–expenditure donors against the fixed SCF balance-sheet vector, so the within-survey dependence is preserved and the companion cross-survey correlations move endogenously; Table B.4 reports them. The result separates by object. Dependence between R and liquid assets X has relatively little effect on the attenuation of the financing-constraint component, which stays between 31.6 and 43.4 percent across the full range from -1 to $+1$. It matters more for *total* attenuation, because it changes how important financing constraints are relative to the ordinary income channel: the same sweep moves total attenuation between 1.1 and 9.1 percent. Dependence between R and capacity K matters a great deal: the same sweep moves the attenuation monotonically from 23.4 percent at a target rank correlation of -0.5 to 61.8 percent at $+0.5$, and further at the endpoints.

Two things follow. The attenuation takes its baseline value only when R and the balance-sheet vector are approximately independent within income groups, which is what conditional random pairing imposes; we report the range rather than the point. And the quantitative uncertainty has two distinct sources that should not be conflated. Dependence observed *within* the SCF is preserved in the preferred baseline by the joint donor, so it is no longer a source of uncertainty here; Tables B.5 and B.6 show what an alternative assignment implies. Dependence *across* surveys, between the CEX financing requirement and the SCF balance-sheet block, is not observed in any dataset, and it is what this sweep varies. If issuers underwrite on income and spending, R and K would be positively dependent and the baseline would understate the attenuation; we regard that as an argument rather than as evidence.

Table B.3 repeats the line-cut experiment of Table 7 across the same targeted dependence constructions. The amplification from a one-percent line contraction stays between 7.2 and 9.3 percent as the target rank correlation between the financing requirement and unused capacity moves from -0.5 to $+0.5$. Unlike the level of the attenuation, this statistic is substantially less sensitive to the assumed dependence, because the same household states enter its numerator and denominator. Table B.4 reports the companion cross-survey correlations that move endogenously in the sweep.

Table B.3: Amplification of the spending response across dependence structures

$\rho_{R,K}$	Rank correlation	Authorized-line contraction		
		−1%	−2.5%	−5%
−0.50		+7.2%	+18.2%	+36.5%
−0.25		+8.4%	+21.2%	+42.6%
0.00		+9.3%	+23.6%	+47.5%
0.25		+9.3%	+23.6%	+47.8%
0.50		+8.6%	+22.0%	+44.7%

Notes: Proportional increase in the modeled short-run spending response, $|\Delta C^\kappa|/|\Delta C^{\text{stable}}| - 1$, under the stated within-income-group rank correlation between the financing requirement and unused capacity. Donors are redrawn to approach each target correlation. The sweep uses an independently simulated set of donor pairings, so its realized $\rho \approx 0$ mean can differ slightly from the preferred baseline mean through simulation alone. Unlike the level of the attenuation, the amplification statistic is substantially less sensitive to the assumed dependence, because the same household states enter its numerator and denominator.

Table B.4: Companion cross-survey correlations in the dependence sweep

Target rank correlation	Realized R, K	Realized R, X	Realized R, A
<i>Targeting Corr(R, K)</i>			
−1.00	−1.000	−0.216	−0.955
−0.50	−0.497	−0.101	−0.474
0.00	−0.014	−0.005	−0.011
0.50	0.469	0.097	0.452
1.00	1.000	0.216	0.955
<i>Targeting Corr(R, X)</i>			
−1.00	−0.216	−1.000	−0.156
−0.50	−0.124	−0.492	−0.092
0.00	−0.014	−0.005	−0.011
0.50	0.090	0.497	0.065
1.00	0.216	1.000	0.156

Notes: Within-income-group weighted rank correlations, averaged over ten simulated pairings. Each exercise reassigns whole CEX income–expenditure donors against the fixed SCF balance-sheet vector, so the within-SCF dependence is preserved and the other cross-survey correlations move endogenously: targeting $\text{Corr}(R, K) = 1$ also drives $\text{Corr}(R, X)$ to 0.216 and $\text{Corr}(R, A)$ to 0.955, because the survey itself links those states. Table B.2 is therefore not a ceteris-paribus experiment in a single correlation. The synthetic units’ (X, A, B) are unchanged across every scenario, which the build asserts.

B.3 Within-SCF dependence and a joint-donor construction

Table B.5: Within-SCF dependence: observed SCF, score-based assignment, and preferred joint donor

States	Pooled rank correlation		Within-income MAD vs SCF		Score-based sign match
	SCF	Score-based	Score-based	Preferred joint	
X, A	0.51	0.35	0.351	0.023	33%
X, B	-0.38	-0.12	0.221	0.052	100%
X, K	0.57	0.34	0.397	0.020	67%
A, u	-0.26	-0.71	0.566	0.036	100%
A, K	0.96	0.98	0.049	0.008	100%
u, K	-0.45	-0.81	0.387	0.049	100%

Notes: X is liquid assets, A the authorized line, B the balance, $u = B/A$, and K unused capacity as in Equation (1). The SCF column joins the main-file credit-limit questions to the liquid-asset measure for the same household-implicate, so it is the dependence actually observed in the survey; it pools the 2016, 2019, and 2022 waves. The middle columns report the mean absolute deviation of the within-income-group weighted rank correlation from its SCF counterpart, for the score-based assignment and for the preferred joint donor used throughout the paper. The last column reports the share of the six income groups in which the score-based assignment reproduces the sign observed in the SCF. That assignment matches each marginal by construction but draws X , A , and u from separate conditional distributions with separate orderings, which weakens $\text{Corr}(X, A)$ and $\text{Corr}(X, K)$ and roughly triples $|\text{Corr}(A, u)|$ relative to the survey.

Table B.6 holds the Diary anchors, the CEX donor and every parameter fixed and varies only how the SCF states are assigned. The preferred joint donor gives 39.2 percent constraint attenuation and 9.5 percent amplification; assignments that do not preserve the within-survey dependence give *larger* attenuation, up to 53.4 percent. Amplification is 9.4 percent with only (A, B) carried jointly and 8.9 percent under the score-based assignment. The latter uses an approximate common price conversion because its pooled line distribution does not retain donor-wave identifiers.

Table B.6: Alternative balance-sheet assignment relative to the preferred joint-donor baseline

Construction	Constr. share	ΔC w/ capacity	Constraint attenuation	Total attenuation	Amplif. of 1% cut
Preferred: joint SCF donor, (X, A, B) from one household	12.6%	-39.87	39.2%	7.1%	9.5%
Joint donor for (A, B) only	12.3%	-39.53	48.3%	9.3%	9.4%
Alternative: separate score- based assignment	11.4%	-39.01	53.4%	10.2%	8.9%

Notes: Quarterly dollars per household at $m = 0.25$. The joint-donor first row is the preferred baseline: one SCF household is drawn per synthetic unit within income-group $\times$ revolver cells, weighted by the SCF weight, and its entire balance-sheet vector is carried, so the within-SCF dependence in Table B.5 is preserved by construction. The last row is an alternative construction, which assigns liquid assets, the line, and utilization from three separate conditional distributions ordered by economic scores; it targets the separate marginals but replaces the dependence the survey observes, and is reported here as an alternative assignment rather than as the baseline. The CEX assignment of income and the financing requirement, λ_x , m , the income shock, and the authorized-line mapping are identical across rows. The three constructions differ only in how the SCF balance-sheet states are assigned. The score-based row uses an approximate pooled price conversion for lines and balances because donor-wave identifiers are unavailable. All three share the same cross-survey pairing draws, and the first row is generated by the same preferred pipeline that produces Table 6 through Table B.8, reproducing the headline numbers in Tables B.1 and 7 exactly.

B.4 What counts as the financing requirement

The contribution of financing constraints to total transmission also depends on what R_i represents. Our baseline uses total CEX expenditure, for the reasons given in Section 6.3. Narrowing the requirement mechanically reduces the share of households for whom the constraint binds.

Table B.7: Sensitivity to the definition of the short-run financing requirement

Financing-requirement definition	Constrained share	Constraint attenuation	Total attenuation	Amplification of a 1% cut
Total expenditure	12.6%	39.2%	7.1%	9.5%
Excluding pension contributions and cash gifts	11.1%	38.1%	5.7%	7.9%
Excluding housing	5.8%	38.2%	3.3%	5.1%
Nondurables and services, net of vehicle purchases	3.5%	32.6%	0.8%	1.3%

Notes: All four concepts are carried by the same CEX donor households, so differences reflect the definition rather than resampling. CEX expenditure aggregates sum the current- and previous-quarter components of the three-month reference period. The narrowest concept is food, apparel, health, entertainment, personal care, reading, tobacco, miscellaneous, and transportation *net of new and used vehicle purchases and vehicle finance charges*, which account for 42 percent of the CEX transportation aggregate. The last column reports the proportional increase in the modeled short-run spending response when authorized open-account lines are cut by one percent, $|\Delta C^{1\%}|/|\Delta C^{\text{stable}}| - 1$, computed on the same households at $m = 0.25$ with the mapping in Equation (8). It is as sensitive to the definition as the level of total attenuation, which is why the headline figure in Section 6.1 is stated for the baseline concept.

Table B.7 reports the consequence. At $m = 0.25$, total attenuation falls from 7.1 percent using total expenditure to 5.7 percent excluding pension contributions and cash gifts, 3.3 percent excluding housing, and 0.8 percent using nondurables and services net of vehicle purchases. The

attenuation of the financing-constraint component moves less but is not constant, from 39.2 percent on total expenditure down to 32.6 percent on the narrowest concept. Two conclusions follow and should not be conflated. Unused card capacity materially relaxes financing constraints under every expenditure concept. Its contribution to *total* transmission is far more sensitive, because the importance of financing constraints depends on how broadly the short-run financing requirement is defined.

The last column of Table B.7 recomputes the same one-percent amplification under each of four concepts: it falls from 9.5 percent on total expenditure to 7.9 percent excluding pension contributions and cash gifts, 5.1 percent excluding housing, and 1.3 percent on nondurables and services net of vehicle purchases. The amplification tracks the level of total attenuation rather than being invariant to it, so the headline figure is a statement about the broad requirement concept. We report it that way throughout and do not select the definition to preserve it.

The financing requirement R_i is measured with CEX total expenditure, summing the current- and previous-quarter components of the three-month reference period, the convention that reproduces published annual aggregates. The narrower concepts in Table B.7 are constructed from the same donor households: *excluding pension contributions and cash gifts* subtracts personal insurance and pensions and cash contributions; *excluding housing* subtracts the housing aggregate; and *nondurables and services* sums food, apparel, health, entertainment, personal care, reading, tobacco, miscellaneous, and transportation expenditure net of new and used vehicle purchases and vehicle finance charges, which together account for 42.2 percent of transport spending in the CEX extract (unweighted). Carrying all four concepts from the same donor ensures that differences across rows reflect the definition rather than resampling.

B.5 The borrowing-cost specification

Section 6.5 adds a behavioral borrowing cap to Equation (5), generated by a convex borrowing cost $\chi_A(b; r) = \eta_1 r b + \frac{\eta_2}{2} b^2$, so that the household borrows $b^* = \min\{\text{gap}, K_i, \bar{B}_i(r)\}$ and desired borrowing falls linearly in the rate. The cost function is a device for writing the cap in a recognizable form, not a microfoundation: it fixes the shape of the cap but not its position, because nothing here specifies the marginal benefit of borrowing that would close the household's problem. We do not estimate η_1 or η_2 and do not claim to identify them. What the exercise does instead is impose an externally calibrated response of desired card borrowing to the APR and report how the answer moves across its plausible range; it is a sensitivity analysis, not an estimate of the borrowing margin, and it does not complete a structural price channel. Under quadratic costs the cap is a dollar level rather than a fraction of the line, $\bar{B}_i = \max\{\theta K_i - \psi \Delta r, 0\}$, and its two terms pose different questions. Write $d \equiv \psi \Delta r$ for the dollar tightening implied by the estimated APR increase, so that ψ remains the structural slope and d the realized amount; ψ is calibrated so that aggregate borrowing falls by a stated percentage, and the percentages we report span the range the interest-elasticity literature supports. The level θK_i

asks whether treating the line as a hard capacity overstates the buffer. The slope ψ asks whether the policy-induced APR increase is itself a transmission channel operating on the borrowing margin. The two have opposite implications for the sign of the net effect, so Table B.8 reports them separately. In the representative household pairing the modeled within-quarter card draw is 1.8 percent of aggregate *unused* capacity—a property of the accounting block, not a measured market aggregate—so the behavioral margin operates on a small base.

Panel A of Table B.8 varies θ and holds the cap at the same level before and after the shock, since a structural reluctance to borrow is a feature of preferences and not of the policy shock. The level of the result is robust. A cap of 70.3 percent of unused capacity still leaves the attenuation of the financing-constraint component at 33.4 percent against 39.2 percent at the line, and total attenuation at 6.0 percent against 7.1 percent.

Panel B instead lets the APR increase tighten the cap, $\bar{B}_i = K_i$ before the shock and $\max\{K_i - d, 0\}$ after, with d calibrated at pre-shock resources so that aggregate borrowing falls by the stated percentage. This is the specification the interest-elasticity literature disciplines, and it requires the calibration to be stated carefully. Gross and Souleles (2002) report that a one percentage-point increase in the account interest rate lowers the debt-to-limit ratio by about 1.7 percentage points in the long run; on their mean limit of \$6,207 and mean debt of \$1,349 that is 7.8 percent of debt per 100 basis points, which their reported elasticity of -1.3 against a mean rate of 16.6 percent approximately reproduces. Scaled to the 1.11 percentage-point APR increase of Fact 1, the implied decline is 8.7 percent; the two-month response in their Figure II gives 5.8 percent. At the long-run anchor the offsetting channel is -40.94 dollars per household per quarter. That more than eliminates the buffer: ΔC becomes $\$ - 80.81$ against $\$ - 42.90$ without cards, so total attenuation is -88.3 percent and the modeled response is *larger* with card capacity than without it.

Table B.8: Soft borrowing-cost specification

Panel A. Behavioral cap present at both the pre- and post-shock APR

Reduction in card borrowing	Implied cap as % of unused capacity	ΔC with capacity	Constraint attenuation	Total attenuation
0.0%	100.0%	-39.87	39.2%	7.1%
1.5%	96.5%	-39.91	38.7%	7.0%
4.0%	91.0%	-39.99	37.7%	6.8%
5.8%	87.3%	-40.05	36.9%	6.6%
8.7%	81.5%	-40.13	35.9%	6.5%
15.0%	70.3%	-40.32	33.4%	6.0%

Panel B. Cap tightened by the APR increase, post-shock only

Reduction in card borrowing	Cap tight. d (\$)	ΔC with capacity	Borrowing-cost component	Total atten.	Agg.-equiv. cut κ	Ampli. of 1% cut
0.0%	0	-39.87	0.00	7.1%	0.0%	+9.49%
1.5%	126	-46.93	-7.06	-9.4%	1.9%	+8.06%
4.0%	355	-58.69	-18.82	-36.8%	4.9%	+6.39%
5.8% [†]	534	-67.16	-27.29	-56.5%	7.1%	+5.55%
8.7% [†]	863	-80.81	-40.94	-88.3%	10.6%	+4.59%
15.0%	1,751	-110.45	-70.57	-157.3%	18.2%	+3.11%

Notes: Quarterly dollars per household at $m = 0.25$; the no-card comparison is $\Delta C = -42.90$, the preferred-baseline value, in both panels. Panel A imposes a behavioral cap $\bar{B}_i = \theta K_i$ in both periods and asks whether the hard-capacity benchmark drives the level of attenuation; θ is calibrated so that post-shock borrowing is the stated percentage below its hard-capacity value. Panel B instead lets the APR increase tighten the cap, $\bar{B}_i = K_i$ before the shock and $\max\{K_i - d, 0\}$ after, where $d = \psi \Delta r$ is the dollar tightening implied by the estimated APR increase, calibrated at pre-shock resources so that aggregate borrowing falls by the stated percentage; this is the specification the interest-elasticity literature disciplines. The aggregate-equivalent column reports the uniform authorized-line cut κ for which $\Delta C^{\text{soft}}(d) = \Delta C^{\text{cut}}(\kappa)$ through Equation (8); that equality is an *aggregate* one and the two experiments need not bind the same households. The last column is this paper's headline statistic recomputed *inside* the soft-cap economy: the amplification of an additional one-percent authorized-line cut, $|\Delta C^{1\%}|/|\Delta C^{\text{soft}}| - 1$, with d held fixed. It is the object the abstract reports, and unlike the total-attenuation column it does not change sign anywhere on the grid: it falls from 9.49 percent at $d = 0$ to 4.59 percent at the Gross and Souleles (2002) long-run anchor and 3.11 percent at the most extreme row. The borrowing-cost channel therefore overturns the *no-capacity comparison* without overturning the counterfactual the paper actually states. Rows marked [†] are the values implied by Gross and Souleles (2002): their Table III implies a 1.7 percentage-point fall in the debt-to-limit ratio per 100 basis points, which on a mean limit of \$6,207 and mean debt of \$1,349 is -7.8 percent per 100 basis points, or -8.7 percent for the 1.11 percentage-point APR increase estimated here; the two-month response in their Figure II gives -5.8 percent. We do not identify the cost parameters from our data.

Three considerations bound how much weight Panel B deserves, and we state them rather than resolve them. First, a behavioral cap and an issuer line cut can deliver the same aggregate spending response without binding the same households. A household with $K_i - d < g_i < K_i$, where g_i is its financing gap, is unconstrained under the hard-capacity benchmark and constrained under the behavioral cap, so the borrowing-cost margin reaches households that were not at their authorized line. The aggregate-equivalent-cut column of Panel B reports the uniform authorized-line cut κ for which $\Delta C^{\text{soft}}(d) = \Delta C^{\text{cut}}(\kappa)$ through Equation (8): an aggregate-equivalent counterfactual, not a household-level equivalence. The two incidences are in fact close here without coinciding. Among households constrained under at least one of the two experiments, 96.9 percent are constrained under both; measured against the full household sample,

0.29 percent are constrained only by the behavioral cap and 0.13 percent only by the equivalent line cut. The aggregate-equivalent-cut column is therefore informative about magnitude. At the long-run anchor the aggregate-equivalent cut is 10.6 percent, an order of magnitude above the 0.8 percent cut that reproduces the no-card *total* response at $m = 0.25$, and far outside the estimated open-account interval in Table B.9. That 0.8 percent cut is not the same object as the $\gamma = -1.07$ percent reported in Appendix B.6, which is stated in the $m = 0$ decomposition where the whole response is the financing-constraint component. Second, the elasticity we are borrowing is a nine- to twelve-month response of the *stock* of revolving debt, whereas the object here is one quarter’s flow of constraint-relieving borrowing; mapping one onto the other one for one is an upper bound on the channel, not a point estimate of it. Third, and following from the non-identification stated at the outset, the calibration fixes d directly rather than η_1/η_2 , so the range reported here is a range over outcomes, not over preference parameters.

B.6 Sensitivity to the stock response

The baseline normalizes the impact response of open-account authorized lines to zero, consistent with Facts 2–5. Table B.9 instead passes the estimated point response and its confidence bounds through the household accounting block using the same authorized-line-to-unused-capacity mapping as Equation (8). The liquidity-buffer effect remains economically meaningful throughout the confidence interval: the attenuation of the financing-constraint component ranges from 20.9 percent at the lower bound to 90.3 percent at the upper bound. In the $m = 0$ decomposition, neutralizing the buffer’s attenuation of the financing-constraint response would require an authorized-line response of $\gamma = -1.07$ percent, 3.07 standard errors below the benchmark estimate. Stated instead on the total response at $m = 0.25$, the cut that reproduces the no-card economy is 0.8 percent; the two are computed at different values of m , which change the constraint-side response and the neutralizing cut. The ordinary component cancels when the two economies are equated at a fixed m . This exercise reflects uncertainty about the stock response, not a second policy shock.

Table B.9: Sensitivity of the accounting exercise to the open-account authorized-line response

γ (ΔA)	Interpretation	Mean ΔK	Constraint attenuation	Total attenuation
-0.50%	lower 95% bound	-0.57%	20.9%	2.7%
+0.00%	baseline normalization	+0.00%	39.2%	7.1%
+0.62%	point estimate	+0.70%	64.2%	13.1%
+1.75%	upper 95% bound	+1.99%	90.3%	23.9%

Notes: γ is an assumed proportional response of open-account authorized lines, applied as $A'_i = (1 + \gamma)A_i$ with $K'_i = \max\{A'_i - B_i, 0\}$, the same mapping as Equation (8). Because unused capacity is a small fraction of the authorized line for high-utilization households, ΔK differs from γ . The baseline row normalizes the impact response to zero, consistent with the near-zero stock estimates in Section 3; the remaining rows use the realized-rate point estimate and its 95 percent confidence bounds. Those bounds are the Student- t interval on 35 residual degrees of freedom reported in Tables 1 and 2, so the first and last rows of this table are the endpoints of the same interval quoted in the text. Neutralizing the modeled attenuation of the financing-constraint response in the $m = 0$ decomposition would require $\gamma = -1.07$ percent, 3.07 standard errors below the point estimate. Computed at $m = 0.25$ for total attenuation and $m = 0$ for the constraint component, on the same Diary anchors and donor rules as Tables B.1, 6 and 7, as means over 50 cross-survey pairings.

The counterfactual is also uniform across the level of the line—the same proportional cut applies to every authorized line—and Section 3 left one pattern unresolved that is not uniform in that sense: under high-frequency identification the open-account estimates are negative only at the upper quantiles, which would be a rotation of the limit distribution rather than the uniform shift that $A'_i = (1 - \kappa)A_i$ imposes. That pattern cuts two ways here, and the two directions should not be run together.

The rotation leaves the baseline approximately intact, but on a premise we state rather than assume. A household enters the financing-constraint channel when unused capacity is thin against what it has to finance, which is the $u \geq 0.80$ group of Table 8, and not when its authorized line is large. To the extent that large authorized lines and high utilization are negatively associated, as they are in the cross-section, capacity trimmed from the top of the limit distribution is capacity removed from households for whom it was slack, and a rotation of the size the high-frequency estimates suggest would leave the modeled constraint component close to what the stable-stock baseline assigns it. That premise is not innocuous. A household can carry a large line and still revolve against most of it, and we have not tested which pattern dominates, because the limit quantiles are account-level in the card panel while utilization enters the household block from a different survey.

The same asymmetry works against the counterfactual. If a synchronized withdrawal of capacity would in fact arrive as a rotation, then a given aggregate withdrawal binds fewer households than the uniform cut of Equation (8) delivers, and the amplification we report is the generous case rather than the conservative one. We state this as a sign and not a magnitude: establishing it would require passing a quantile-dependent cut through the household block, and the card-market quantiles are account-level while the household file is not, so the two cannot be linked at the household level to support it. The honest summary is that the uniform mapping is a simplification whose error runs toward the paper’s conclusion on the counterfactual and away

from it on the baseline.

One margin deliberately stays outside the quantitative block. Fact 5 finds no decline in the incidence of line increases, but incidence is not size: the series records how many accounts receive an increase, not how large the increases are, so it cannot identify the dollar value of any increases that were forgone. Mapping a change in incidence into $A'_i = (1 - \kappa)A_i$ would require both objects. The baseline therefore continues to set $A'_i = A_i$, which is a statement about capacity already authorized at the moment of the shock and remains valid whether or not tightening slows the subsequent growth of lines.

B.7 Horizon and incidence

Pre-authorized capacity is most valuable on impact because it is already in place. A household without a card can eventually substitute toward liquid savings, but doing so takes time. We discipline the adjustment with CEX saving capacity, measured as after-tax income less total expenditure within income groups, and allow the no-card economy to rebuild liquid assets over a horizon H up to the capacity it lost. Because the rebuilding path is imposed rather than chosen, this is a scenario rather than model-generated dynamics, and Figure B.1 reports it. One feature of it does not depend on the timing assumption: measured saving rates are negative in the two lowest income groups, so the households most likely to be financing constrained are also those least able to replace a lost line through saving.

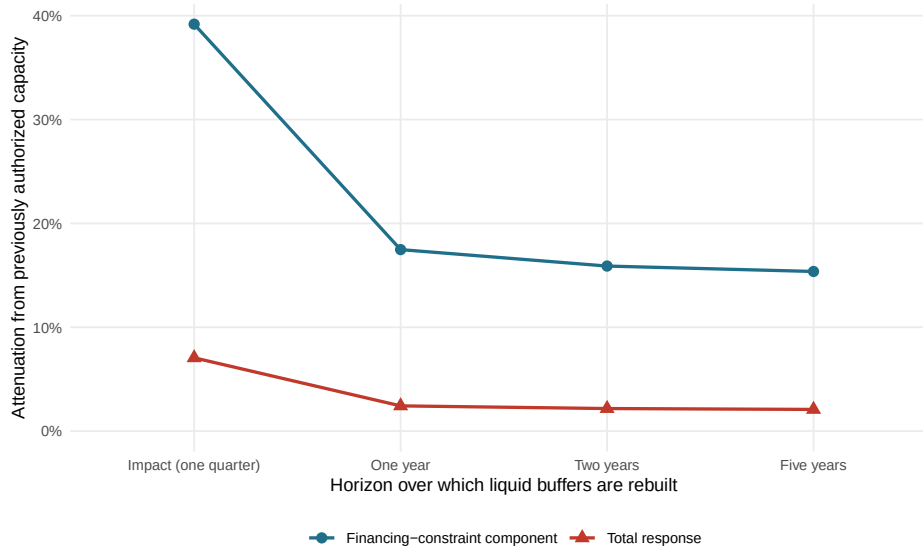


Figure B.1: Attenuation under a mechanical liquid-asset rebuilding scenario

Notes: The rebuilding path is imposed from CEX saving capacity (after-tax income less total expenditure) by income group, not generated by household optimization. It is a scenario, not a dynamic forecast. On impact the attenuation of the constraint component is 39.2 percent and of the total response 7.1 percent. After one year of rebuilding these fall to 17.5 and 2.4 percent; after two years to 15.9 and 2.2 percent; after five years to 15.4 and 2.1 percent. Measured saving rates are negative in the two lowest income groups, which are assigned zero rebuilding capacity.

Capacity is also unevenly distributed, and that descriptive fact is not in doubt: card access rises from 47.3 percent of households below \$25,000 to 97.7 percent above \$150,000, and median unused capacity among cardholders rises from \$2,679 to \$44,196, both directly from the Diary and the SCF. The modeled incidence of the financing-constraint component, by contrast, does depend on the cross-survey match: the same R - K dependence that moves the aggregate attenuation also moves its distribution across income groups, so we report those shares in Table B.10. We make no claim about the incidence of the total consumption response, since under a common ordinary MPC the income channel is distributed with income by construction.

Table B.10: Incidence of the financing-constraint component

Household income	Without K	With K	Difference
Under 25,000	11.7%	15.7%	4.0
25,000–49,999	27.4%	30.7%	3.3
50,000–74,999	19.9%	20.4%	0.5
75,000–99,999	10.0%	9.2%	–0.8
100,000–149,999	18.5%	14.7%	–3.9
150,000+	12.4%	9.4%	–3.1

Notes: Shares of the aggregate financing-constraint component of the consumption response, in percent. These shares do not describe the incidence of the total consumption response: under a common ordinary MPC the income channel is distributed with income by construction.

These shares inherit the cross-survey dependence structure examined in Appendix B.2 and should be read alongside it.

B.8 Sources of statistical uncertainty

Table B.11: Sources of statistical uncertainty in the household block

Statistic	Estimate	Standard error from			
		Pairing	Imputation	CEX	Combined
Constrained share (%)	12.60	0.079	0.295	0.255	0.398
ΔC , stable stock (\$)	-39.87	0.075	0.062	0.264	0.281
ΔC , 1% line cut (\$)	-43.66	0.127	0.116	0.351	0.391
Amplification of a 1% cut (%)	9.49	0.175	0.170	0.379	0.451
Attenuation, constraint component (%)	39.20	0.461	1.003	0.512	1.217
Attenuation, total (%)	7.06	0.099	0.245	0.183	0.321

Notes: All three components are computed on the same accounting engine, constants and seeds as Table B.1. *Pairing* is the Monte Carlo standard error of the reported mean across the 50 cross-survey pairings, $\text{sd}/\sqrt{50}$; it is the component most often reported on its own. *Imputation* is the standard deviation across five runs in which the SCF donor pool is restricted to a single implicate; the baseline pools all five with the public weight divided by five, so the point estimate already averages over imputation and this is the between-implicate spread around it. *CEX* is a balanced repeated replication standard error over the 44 **WTREP** replicate weights supplied with the Interview survey, applied to the donor-selection probabilities; $V = \frac{1}{44} \sum_r (\hat{\theta}_r - \theta_0)^2$. *Combined* adds the three in quadrature. The Diary and SCF replicate weights are not applied and the components are treated as independent, so the reported uncertainty is conditional on those simplifications. The relative size of statistical uncertainty and sensitivity to assumptions differs by statistic; the text compares amplification and constraint attenuation separately.

B.9 Liquid assets and the price base

Table B.12 varies the usable share of liquid assets λ_x , which the baseline sets to one. Lowering it raises the constrained share in both economies and raises the attenuation of the financing-constraint component, so $\lambda_x = 1$ is the conservative choice for our claim.

Table B.12: Sensitivity to the usable share of liquid assets

λ_x	Constrained with K	Constrained without K	ΔC with K	ΔC without K	Constraint attenuation
0.25	16.0%	24.8%	-8.65	-15.97	45.8%
0.50	14.5%	21.6%	-7.58	-13.31	43.1%
0.75	13.4%	19.5%	-6.85	-11.58	40.8%
1.00	12.6%	17.9%	-6.28	-10.33	39.2%

Notes: Computed at $m = 0$ so that the entire response is the financing-constraint component, using total expenditure as the financing requirement. The baseline sets $\lambda_x = 1$; that row reproduces the preferred baseline exactly, which the build asserts. Aggregates are means over the same 50 cross-survey pairings used throughout.

The preferred household block uses a common 2024 price base; the pooled-SCF matching-score alternative in Table B.6 uses the approximate conversion disclosed above. Table B.13 reports the alternative in which the balance-sheet block and the matching-cell income sit on

different bases, and shows that the alternative yields a smaller constrained share, smaller attenuation and amplification percentages, and a smaller absolute spending response in that comparison.

Table B.13: Price base of the household block

Price base	Constr. share	ΔC w/ capacity	Constraint attenuation	Total attenuation	Amplif. of 1% cut
Common 2024 base (baseline)	12.6%	−39.87	39.2%	7.1%	9.5%
Survey-year nominal, X in 2022 dollars	12.1%	−39.28	38.4%	6.4%	8.6%

Notes: Quarterly dollars per household at $m = 0.25$, on the same Diary anchors, donor rules and cross-survey pairing draws; the rows differ only in the price base. The first row is the baseline: the SCF block is carried to 2024 dollars with the Board’s per-wave Summary Extract factor and then CPIAUCSL from 2022 to 2024, the CEX by interview quarter, and the income cells are cut on the reflatd income. The second row is an alternative convention in which liquid assets carried the Board’s adjustment to 2022 dollars while the authorized line, the balance and the matching-cell income stayed in the survey-year nominal dollars of the pooled 2016, 2019 and 2022 waves, so that within a single donor household the two sides sat on different bases. Reflation works through two channels. Rescaling levels alone raises unused capacity against an unchanged financing requirement and lowers the constrained share; but it also moves a large share of the weighted donor pool into a different income cell, and that recomposition dominates. The alternative yields smaller shares and attenuation/amplification percentages, and a smaller absolute spending response. Card-market limits, which enter only the regressions of Section 3, are deflated by CPIAUCSL throughout and are unaffected.

References

- Agarwal, S., Chomsisengphet, S., Mahoney, N., and Stroebel, J. (2018). Do banks pass through credit expansions to consumers who want to borrow? *Quarterly Journal of Economics*, 133(1):129–190.
- Argudo, E. P. (2025). Monetary policy transmission via nonbank lending: Evidence from peer-to-peer loans. *Journal of Financial Stability*, 80:101455.
- Auclert, A. (2019). Monetary policy and the redistribution channel. *American Economic Review*, 109(6):2333–2367.
- Aydin, D. (2022). Consumption response to credit expansions: Evidence from experimental assignment of 45,307 credit lines. *American Economic Review*, 112(1):1–40.
- Baker, S. R. (2018). Debt and the response to household income shocks: Validation and application of linked financial account data. *Journal of Political Economy*, 126(4):1504–1557.
- Baker, S. R., Farrokhnia, R. A., Meyer, S., Pagel, M., and Yannelis, C. (2023). Income, liquidity, and the consumption response to the 2020 economic stimulus payments. *Review of Finance*, 27(6):2271–2304.
- Bauer, M. D. and Swanson, E. T. (2023). A reassessment of monetary policy surprises and high-frequency identification. *NBER Macroeconomics Annual*, 37:87–155.
- Bord, V. M., Kovacs, A., and Moran, P. (2026). Automated credit limit increases and consumer welfare. *Journal of Monetary Economics*, 159:103901.
- Bosshardt, J., Di Maggio, M., Kakhbod, A., and Kermani, A. (2024). The credit supply channel of monetary policy tightening and its distributional impacts. *Journal of Financial Economics*, 160:103914.
- Bräuning, F. and Stavins, J. (2025). The credit card spending channel of monetary policy: Micro evidence from account-level data. Research Department Working Paper 25-10, Federal Reserve Bank of Boston.
- Braxton, J. C., Herkenhoff, K. F., and Phillips, G. M. (2024). Can the unemployed borrow? Implications for public insurance. *Journal of Political Economy*, 132(9):3025–3076.
- Chava, S., Ganduri, R., Paradkar, N., and Zeng, L. (2023). Shocked by bank funding shocks: Evidence from consumer credit cards. *Review of Financial Studies*, 36(10):3906–3952.
- Cloyne, J., Ferreira, C., and Surico, P. (2020). Monetary policy when households have debt: New evidence on the transmission mechanism. *Review of Economic Studies*, 87(1):102–129.

- Cucic, D. and Gorea, D. (2026). Nonbank lending and the transmission of monetary policy. *Review of Financial Studies*, 39(4):966–1014.
- Drechsler, I., Jung, H., Peng, W., Supera, D., and Zhou, G. (2025). Credit card banking. Staff Report 1143, Federal Reserve Bank of New York.
- Drechsler, I., Savov, A., and Schnabl, P. (2017). The deposits channel of monetary policy. *Quarterly Journal of Economics*, 132(4):1819–1876.
- Fulford, S. L. (2015). How important is variability in consumer credit limits? *Journal of Monetary Economics*, 72:42–63.
- Fulford, S. L. and Schuh, S. D. (2023). Revolving versus convenience use of credit cards: Evidence from U.S. credit bureau data. *Journal of Money, Credit and Banking*, 55:1667–1701.
- Furceri, D., Mazzola, F., and Pizzuto, P. (2019). Asymmetric effects of monetary policy shocks across U.S. states. *Papers in Regional Science*, 98(5):1861–1891.
- Fuster, A., Kaplan, G., and Zafar, B. (2021). What would you do with \$500? Spending responses to gains, losses, news, and loans. *Review of Economic Studies*, 88(4):1760–1795.
- Grigoli, F. and Sandri, D. (2026). Monetary policy and credit card spending. *European Economic Review*, 185:105294.
- Gross, D. B. and Souleles, N. S. (2002). Do liquidity constraints and interest rates matter for consumer behavior? *Quarterly Journal of Economics*, 117(1):149–185.
- Gross, T., Notowidigdo, M. J., and Wang, J. (2020). The marginal propensity to consume over the business cycle. *American Economic Journal: Macroeconomics*, 12(2):351–384.
- Herkenhoff, K. F. (2019). The impact of consumer credit access on unemployment. *Review of Economic Studies*, 86(6):2605–2642.
- Jarociński, M. and Karadi, P. (2020). Deconstructing monetary policy surprises—the role of information shocks. *American Economic Journal: Macroeconomics*, 12(2):1–43.
- Johnson, D. S., Parker, J. A., and Souleles, N. S. (2006). Household expenditure and the income tax rebates of 2001. *American Economic Review*, 96(5):1589–1610.
- Jordà, O. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182.
- Kaplan, G., Moll, B., and Violante, G. L. (2018). Monetary policy according to HANK. *American Economic Review*, 108(3):697–743.

- Kaplan, G., Violante, G. L., and Weidner, J. (2014). The wealthy hand-to-mouth. *Brookings Papers on Economic Activity*, 45(1):77–138.
- La Cava, G., Hughson, H., and Kaplan, G. (2016). The household cash flow channel of monetary policy. Research Discussion Paper 2016-12, Reserve Bank of Australia.
- Littlejohn, M. (2025). The credit card and small business lending channels of monetary policy. *Journal of Economic Dynamics and Control*, 171:105045.
- Parker, J. A., Souleles, N. S., Johnson, D. S., and McClelland, R. (2013). Consumer spending and the economic stimulus payments of 2008. *American Economic Review*, 103(6):2530–2553.
- Santucci, L. (2015). A tale of two vintages: Credit limit management before and after the CARD Act and Great Recession. Payment Cards Center Discussion Paper DP 15-01, Federal Reserve Bank of Philadelphia.
- Sofianos, G., Melnik, A., and Wachtel, P. (1990). Loan commitments and monetary policy. *Journal of Banking and Finance*, 14(4):677–689.
- Woodford, M. (1996). Loan commitments and optimal monetary policy. *Journal of Monetary Economics*, 37(3):573–605.